



**9349 ITE 17 Data Mining Final Project**

**Mining the Literacy Gap: Predictive Analytics on the Impact of Early Grade Proficiency,  
and Regional Disparities on Overall Student Performance in the Philippines**

**RESERVED**

Aguilan, Jecquar Aguilan

Angara, Mark

Barrera, Ryan Adelard

Karim, Syed

Padiernos, Leon Neil

Tibayan, Gelina

Torres, Daniel Linus

Vergara, Marvin John

Wacdagan, Benedict

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Mrs. Beverly Ferrer

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### **ABSTRACT**

The impact of MT (L1) proficiency on L2 (Filipino) and L3 (English) proficiency was investigated by analyzing the school-level CRLA 2025-2026 data of 5 regions encompassing 6,121 elementary schools and 647,681 students. To explore the extent to which L1 foundations relate to L2 and L3 achievements and whether there is any relation variability based on regional differences and school categories, correlational analysis, and Logistic Regression were applied in the research. The obtained correlations from the correlational analysis at Grade 3 showed high and significant correlations between L1 and L2 from  $r = .9162$  between Mother Tongue and Filipino to  $r = .8074$  between Mother Tongue and English, which led to the rejection of the null hypothesis for Grade 3. However, there were also significant differences across regions: Region III (Central Luzon) had the highest At Grade Level (18.40%) and Region VIII (Eastern Visayas) the lowest (12.69%). Logistic Regression showed a relatively low prediction rate (from 56% to 68%) and indicated that the achievement level in schools cannot be guaranteed by demographic and linguistic attributes. This research then confirmed the assumption of cross-linguistic transfer effect but suggested that the quality of implementation and region become important.

Keywords: Mother Tongue-Based Multilingual Education (MTB-MLE), language of instruction, literacy assessment, data mining, correlational analysis, Philippine elementary education, Republic Act No. 12027, CRLA



TABLE OF CONTENTS

**ABSTRACT.....2**

**INTRODUCTION..... 4**

    Background of the Study..... 4

    Research Questions..... 8

    Significance of the Study..... 9

    Hypotheses of the Study..... 9

**METHODOLOGY.....11**

    Research Design..... 11

    Summary Table of Results and Discussion to the Research Questions..... 11

    Data Collection..... 12

    Dataset Attributes.....13

    Data Preprocessing.....17

    Modeling..... 18

**RESULTS AND DISCUSSION.....20**

    Demographic and Geographic Profile of Participating Schools..... 21

    Status of Early-Grade Literacy per Region and Classification..... 28

    Correlation Between Mother Tongue, Filipino, and English Proficiency..... 34

    Distinct Performance Clusters Emerge from Grade 1–3 Reading Trajectories..... 40

    Predictive Modeling: Determinants of School-Level Reading Success..... 46

**CONCLUSION..... 62**

**REFERENCES.....67**

**APPENDICES..... 74**



## **INTRODUCTION**

### **Background of the Study**

The educational system of the Philippines is geared towards making sure that learners are taken seamlessly from the stage of foundation schooling to their chosen career path. It operates through the direction of four entities: the Department of Education (DepEd), which handles basic education using the K–12 system, ranging from kindergarten up to elementary, junior, and senior high school. Those who complete K–12 education may then join higher education institutions that have been accredited by the Commission on Higher Education (CHED). For learners who go through technical training, the appropriate training offered by TESDA will be considered. As for professions needing licensure examinations, the Professional Regulation Commission (PRC) conducts these examinations. (The Philippine Education System: Complete Guide to CHED, TESDA, DEPED, and PRC, 2026.)

One component included in the K-12 system was the application of Mother Tongue Based Multilingual Education (MTB-MLE). MTB-MLE became a mandatory practice following the passage of Republic Act No. 10533, which is the Enhanced Basic Education Act of 2013. Section 4 requires that for Kindergarten and Grades 1 to 3, the instruction, learning materials, and evaluation process must be done using the mother tongue of the learner. In addition, the Department of Education (DepEd) must develop a mother language transition program for Grades 4 to 6 wherein the learners would gradually transition to Filipino and English as the language of instruction prior to secondary education. Furthermore, Section 5(f) of the same law mandates that the curriculum must utilize the principles of MTB-MLE starting with the existing knowledge base of the learners provided that there are learning materials and competent teachers (Republic Act No. 10533, 2013).

In order to put this legal mandate into effect, DepEd Order No. 16, s. 2012 was issued, which provides specific guidelines on the national implementation of MTB-MLE starting SY 2012-2013. It mandates that the use of the mother tongue be adopted as the medium of instruction in all public schools from Kinder up to Grade 3, and the main goal is the realization



of “Every Child-A-Reader and A-Writer by Grade 1.” The said order identifies twelve main languages, which include Tagalog, Cebuano, Iloko, Hiligaynon, and Waray, as languages to be offered as learning areas and used as the medium of instruction. This policy is based on the pilot projects *Lingua Franca* (1999-2001) and *Lubuagan* (1999-present), which produced valuable information about the possible benefits of mother tongue-based education. Also, more than nine hundred schools, including indigenous communities, have implemented MTB-MLE with the assistance of different organizations such as BEAM, TAP, and SIL (Department of Education, 2012).

Implementation of the policy at the school level entailed several operational steps. Schools were supposed to carry out language mapping, identify the mother tongue of each learner, group them according to their mother tongue, and organize the classes in accordance with the linguistic makeup of the learners. For schools having learners with multiple languages, the class models were adjusted in accordance with the linguistic environment: monolingual classes were used for schools where one language was prominent; if there were two or more languages, schools were supposed to adopt bi- or trilingual classes, and at times sought help from the parents who were natives of the languages of the minorities. The deployment of the teachers also needed to take into account linguistic matches, whereby the teachers were supposed to be the native speakers of the language spoken by the learners. Finally, all learning materials and evaluation tools needed to be localized and developed in the mother tongue, after being subjected to a quality assurance process in the school, division, or region level (Department of Education Region II, 2020).

The basis for this concept goes back before MTB-MLE itself. Cummins's (1979) linguistic interdependence hypothesis proposes that a learner's competencies in the first language—such as decoding skills, vocabulary knowledge, and comprehension strategies—are not confined to that language but transfer to the acquisition of a second and third language once a sufficient threshold of proficiency in the mother tongue has been reached. This hypothesis is grounded on the concept of cross-linguistic transfer, where an individual with good command over their first language (L1) will be able to acquire other languages such as Filipino (L2) and



English (L3) with ease. This concept is empirically confirmed by a study conducted at the Philippine Institute for Development Studies (PIDS, 2025). According to the results obtained by the researchers, children perform better not only in L1 but in Filipino, English, and Mathematics too when learning in the language they comprehend. Moreover, the positive effects can reach one additional year of schooling and increase completion rates by 9 to 12 percentage points from Grades 1 through 6. Linguistic mismatch is referred to by the researchers as "the tax on learning" since each additional unit of mismatch reduces achievement by 35 to 57 percent of potential performance, while the MTB-MLE policy reduces such losses by two-thirds across the country (Abrigo et al., 2025).

To test the effectiveness of MTB-MLE, the Department of Education, along with the ABC+ Project of UNICEF, rolled out Regional Memorandum No. 000608, s. 2021, which required all Schools Division Offices to submit literacy data on first key stage learners in the Mother Tongue-Based learning area. The memorandum states that both the Comprehensive Rapid Literacy Assessment (CRLA) and the Early Grade Reading Assessment (EGRA) shall be used as tools in assessing reading literacy in the Mother Tongue-Based learning area. The data gathered from these assessments would serve as the basis for selecting priority divisions and schools to be included in the rollout of the CRLA under the project. This assessment framework gave DepEd the capacity to monitor the theoretical promise of MTB-MLE; such that strong L1 skills transfer onto L2 and L3 acquisition, and whether it was actually materializing in classrooms, which would provide empirical evidence to inform policy decisions and instructional interventions (Department of Education, 2021).

Literacy outcomes in the Philippine elementary schools are also shaped by factors that operate independently of the language of instruction. According to Kilag et al. (2024), the reading abilities of the learner are influenced by financial difficulties, the availability of an appropriate number of reading materials, teacher training, and the involvement of parents. The learners from poor backgrounds generally exhibit less proficiency in literacy skills regardless of the language of instruction. Undernutrition intensifies the problem, as evidenced by the link between undernutrition and the learning crisis in the country according to EDCOM 2. Many



Filipinos carry the burden of illiteracy owing to poor nutrition from childhood years (EDCOM 2 Communications, 2026). Because the CRLA dataset used in this study does not capture household income, health status, or nutrition directly, these factors are acknowledged here as plausible influences on literacy that may operate alongside, or independently of, mother tongue matching.

This issue has gained significance through the enactment of a drastic change in policy. More than a decade since the MTB-MLE initiative was introduced, the legislative body reached the conclusion that most educational institutions were never able to provide it due to the lack of enough instructors and resources related to mother tongue instructions in classrooms (EDCOM 2 Communications, 2024). As a result, Republic Act No. 12027 was passed, called the Act Discontinuing the Use of the Mother Tongue as Medium of Instruction from Kindergarten to Grade 3, which came into effect on October 11, 2024, amending Sections 4 and 5 of Republic Act No. 10533. According to the new legislation, the use of any other language as a mediator language will be optional in classrooms taught in one language and meeting specified spelling and resource requirements (Republic Act No. 12027, 2024). Despite the empirical evidence provided by Abrigo et al. (2025), that language matching is related to academic success, the new legislation can be regarded as a dismissal of the theoretical concept of MTB-MLE.

But some considerations have to be taken into account during the analysis of the data set, which are: early grade proficiency, school resources and regional disparity. Early grade proficiency is crucial in terms of successful academic performance. As it was stated by Casido (2024), early literacy skills including listening comprehension, letter-sound recognition, initial sound recognition, and familiar word reading are important factors of academic success. School resources matter as well because the introduction of MTB-MLE demanded certain capabilities that include the language map creation, appropriate teachers' assignment based on the learners' native languages and local learning materials (Second Congressional Commission on Education, 2026). Nevertheless, the Philippine educational system suffers from persistent shortage of infrastructure and teachers. Regional disparity matters as well, since poverty and human development differ significantly from one region to another region, especially comparing the



National Capital Region to the Bangsamoro Autonomous Region in Muslim Mindanao (*BTI 2026 Philippines Country Report*, 2026). It could be said that these factors hinder the effectiveness of mother tongue instruction due to the lack of necessary learning materials and experienced teachers.

The purpose of this paper is to assess the effect of early grade mother tongue proficiency and regional disparities on Filipino and English proficiency while the policy was in place. In this context, proficiency is defined as the foundational mastery of early literacy skills, which encompasses listening comprehension, letter-sound recognition, initial sound recognition, and familiar word reading (Casido, 2024). The basis of the policy was that teaching in the first language facilitates acquisition of both Filipino and English languages. Given the sudden reversal of policy without any empirical research conducted, there is a need to examine whether L1-based instruction actually translated into measurable improvements in Filipino and English proficiency during the policy's implementation.

### **Research Questions**

This paper was designed specifically sought to answer the following questions:

1. What is the demographic and geographic profile of the participating schools in terms of:
  - 1.1. Region and Mother Tongue; and
  - 1.2. School Classification (Urban, Partially Urban, Rural)
2. What do early grade students' percentage distributed across five levels (Low Emerging, High Emerging, Developing, Transitioning, At Grade Level) of early grade reading skills level across Region and Classification?
3. Is there a significant association between L1 proficiency and L2, and L3 and whether the association is significantly different across regions, school type (Urban/Rural) or the primary language of school instruction grouping?
4. How many natural clusters of schools can be individually distinguished based on the early grade (Grades 1-3) reading trajectories data through a hierarchical clustering technique?



5. What is the school's probability of achieving majority high proficiency in Filipino (L2) and English (L3) given the schools' region, being in Urban or Rural areas, having Filipino or English as Mother Tongue of the students, having a large or small number of teachers, and a large or small student population in the school?

### **Significance of the Study**

This study could be beneficial to different individuals and institutions related to Philippines basic education. First, the results would be very helpful to policymakers and lawmakers. The study is conducted at a critical period, as the new educational policy (Republic Act No. 12027) was just approved in 2024. The act cancels the necessity to teach in the learner's mother tongue. The act discontinued the mandatory use of the mother tongue primarily due to practical implementation challenges rather than conclusive evidence disproving the hypothesis of cross-linguistic transfer.

Secondly, the results of the study provide numbers that would allow shaping curriculum and language transition programs. Understanding the degree of influence of mother tongue foundations on future language competence is important for the Department of Education as it will help to adjust the multilingual education system and analyze its weak and strong sides. Moreover, school administrators and teachers will also have a chance to benefit from the results of the research and apply them to teaching techniques and resource allocation, particularly in schools with diverse languages, where the MTB-MLE program was difficult to implement.

Lastly, the research will add to the limited evidence base related to cross-linguistic transfer in the Philippine context. It will serve as a ground for further studies of various factors influencing language competence and academic success of learners.

### **Hypotheses of the Study**

***Null Hypothesis ( $H_0$ ):*** At a school level, there is no significant correlation between student proficiency in Mother Tongue (L1) and in Filipino (L2) and in English (L3).

***Alternative Hypothesis ( $H_a$ ):*** The school level student proficiency in Mother Tongue (L1) is significantly related to the student proficiency in Filipino (L2) and English (L3).



## **Scope and Limitations**

In this research, the analysis of the data from schools was conducted to find out whether there is any connection between the level of students' mastery of Mother Tongue (L1) and the outcomes in Filipino (L2) and English (L3) among early grade students in the Philippines. Correlation and predictive methods was used to analyze the existing student language performance data since the beginning of the implementation of the MTB-MLE policy starting from School Year 2012-2013 to check the existence and strength of such connection and the concept of cross-linguistic transfer.

The study is accompanied by a few limitations. First of all, the research is bound to the use of existing data. Thus, the dataset may not contain some information that would be needed for a more complete picture. Some variables like family income, health, and nutrition that have a known impact on literacy may not be included into the dataset, thus, influencing its completeness. Second, this research measures correlations, but not the causal effect. This means that even if there was a strong correlation between L1 performance and L2/L3 outcomes, it is not possible to speak about causality since there are many other reasons that could contribute to the result. Third, the findings of the research will not be applicable to all Philippine schools since in the areas where the number of languages was higher and the MTB-MLE policy had more challenges in terms of lack of teaching material and insufficient preparation of teachers, there might be no sufficient data. Also, there may be problems with data completeness and accuracy that can influence the results of the research. Fourth, the results of this analysis are limited to the period of the implementation of the MTB-MLE policy.



METHODOLOGY

The design of this study is quantitative, non-experimental, correlational research designed to determine the relationship between citizens' mother tongue (L1) and their literacy performance in Filipino and English in various regions of the Philippines. Secondary data from the Comprehensive Rapid Literacy Assessment (CRLA) is used to examine if there are differences in literacy outcomes among elementary students who are taught in the language of the mother tongue at the mother-tongue level. The data collection process, data preprocessing procedures, and the modeling approach used to answer the study's research questions are described below.

Research Design

A quantitative correlational-comparative design is used for the study. The design is appropriate because this study is not experimental but a comparison between the groups of participants in terms of differences in dependent variables (literacy outcomes) of the study based on the independent variables /grouping variables (mother tongue, region, and school locale). This helps them learn to identify statistically significant patterns and associations between mother tongue and literacy attainment in Filipino and English, but have no causal relationship drawn between the two (Ghanad, 2023). This technique allows the researchers to view statistically significant patterns and associations between mother tongue and literacy attainment in Filipino and English, without inferring causality.

Summary Table of Results and Discussion to the Research Questions

| Research Question                                                                                                                                                                                                                  | Page         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <b>RQ1:</b> What is the demographic and geographic profile of the participating schools in terms of:<br><br>1.1. Region and Mother Tongue; and<br>1.2. School Classification (Urban, Partially Urban, Rural)                       | <b>21-27</b> |
| <b>RQ2:</b> What do early grade students’ percentage distributed across five levels (Low Emerging, High Emerging, Developing, Transitioning, At Grade Level) of early grade reading skills level across Region and Classification? | <b>28-33</b> |
| <b>RQ3:</b> Is there a significant association between L1 proficiency and L2, and L3 and whether the association is significantly different across                                                                                 | <b>34-40</b> |



|                                                                                                                                                                                                                                                                                                                                                 |              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| regions, school type (Urban/Rural) or the primary language of school instruction grouping?                                                                                                                                                                                                                                                      |              |
| <b>RQ4:</b> How many natural clusters of schools can be individually distinguished based on the early grade (Grades 1-3) reading trajectories data through a hierarchical clustering technique?                                                                                                                                                 | <b>40-45</b> |
| <b>RQ5:</b> What is the school’s probability of achieving majority high proficiency in Filipino (L2) and English (L3) given the schools' region, being in Urban or Rural areas, having Filipino or English as Mother Tongue of the students, having a large or small number of teachers, and a large or small student population in the school? | <b>45-61</b> |

Table 1. Summary Table of Results and Discussion to the Research Questions

Table 1 maps each Research Question (RQ) to the specific data mining method used to answer it.

Data Collection

The study CRLA 2025-2026 (Comprehensive Rapid Literacy Assessment) is composed of secondary data available at the school level from the Department of Education (DepEd).6,121 rows-one row for each public elementary school, with 58 columns defining school characteristics like the geographical location of the school, its primary language of instruction, and the percentage of students in Grades 1 to 3 who can read.

Researchers enhanced the CRLA data by merging it with three others from the DepEd, such as the School-Personnel-in-SY-2023-2024 data (which contains the total count of teachers at each school), School Facilities in SY 2023-2024 (which shows the amount of facilities present at school), and Enrollment-in-SY-2024-2025 data (which details the overall number of enrollees per school). Additionally, the researchers incorporated the Early Language Literacy and Numeracy Assessment (ELLNA) data, which provides the overall Mean Percentage Score (MPS) per school to get a more accurate perspective on their school resources, demographics, and academic outputs.

Following a merge of the CRLA and a comprehensive listing of schools by the DepEd Masterlist of Schools to capture more specific identifiers and geo-locators for schools(e.g., region, division, municipality, district, barangay)-which were missing in the CRLA-raw file-the added information was matched with the existing data using either a shared School ID number or



standardized school name. Researchers used the merged dataset to evaluate the associations between early-grade literacy outcomes and schools' overall school resources and performance metrics.

The study spans 44 school divisions in five regions (CAR, Region I, Region III, Region VIII and Region IX) across 13 mother tongues (including Tagalog, Waray, and Ilocano), which offers a robust dataset with linguistic and geographic variation to support the comparison of learner literacy outcomes from diverse language backgrounds, as asked in the research question.

Note that the research did not involve collecting new information but utilized previously gathered, organized, and already standardized data from DepEd as part of the CRLA implementation process. Researchers have downloaded the needed files, combined the CRLA and Masterlist, checked for data integrity, and noted down all variables for the subsequent data analyses.

**Dataset Attributes**

There are 58 attributes within the dataset, and can be grouped into four areas: (1) “School and Geographic Descriptors” – these are variables that identify the location of the school; (2) “School Descriptors” – this group identifies various school-specific attributes; (3) “School-wide literacy measures” – this groups indicates how the school performed in literacy overall; (4) “School-specific literacy measures” – this group allows us to see the performance in grade-level and by language performance. The summary of the important attributes in the dataset is below in Table 2.

| <i>Attribute</i> | <i>Data Type</i>      | <i>Description</i>                                                                                   |
|------------------|-----------------------|------------------------------------------------------------------------------------------------------|
| region           | Categorical (nominal) | Administrative region of the school<br><br>(e.g., CAR, Region I, Region III, Region VIII, Region IX) |



|                                                                                         |                       |                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| division                                                                                | Categorical (nominal) | Schools Division Office / province<br><br>under the region (44 unique<br><br>divisions)                                                                                                                               |
| language                                                                                | Categorical (nominal) | Mother tongue (L1) used for<br><br>instruction and assessment at the<br><br>school (e.g., Tagalog, Waray,<br><br>Sinugbuanong Binisaya, Ilocano,<br><br>Pangasinan, Kapampangan, and<br><br>other minority languages) |
| Municipality,<br><br>District_Name,<br><br>Barangay, Province,<br><br>District, Address | Categorical (nominal) | Geographic identifiers that determine<br><br>and help in locating the school                                                                                                                                          |
| School_Name                                                                             | Categorical (nominal) | The names of the school                                                                                                                                                                                               |
| total_enrollees                                                                         | Numeric (discrete)    | The numbers of enrollees for the<br><br>particular school                                                                                                                                                             |
| total_facilities                                                                        | Numeric (discrete)    | Total numbers of school facilities for<br><br>the particular school                                                                                                                                                   |
| total_teachers                                                                          | Numeric (discrete)    | Total number of teachers for the<br><br>particular school                                                                                                                                                             |
| n_test_takers_y                                                                         | Numeric (discrete)    | The total number of students who<br><br>took the Early Language, Literacy,<br><br>and Numeracy Assessment (ELLNA)                                                                                                     |



|                   |                       |                                                                                                                                                       |
|-------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| english_mps       | Numeric (continuous)  | The Mean Percentage Score (MPS) achieved by students in the English subject assessment                                                                |
| filipino_mps      | Numeric (continuous)  | The Mean Percentage Score (MPS) achieved by students in the Filipino subject assessment                                                               |
| mother_tongue_mps | Numeric (continuous)  | The Mean Percentage Score (MPS) achieved by students in the Mother Tongue (L1) subject assessment                                                     |
| numeracy_mps      | Numeric (continuous)  | The Mean Percentage Score (MPS) achieved by students in the Numeracy/Mathematics assessment                                                           |
| overall_mps       | Numeric (continuous)  | The overall or composite Mean Percentage Score across all assessed subjects, serving as a general summary measure of school-wide academic performance |
| Ownership         | Categorical (nominal) | School ownership type (Public)                                                                                                                        |
| Classification    | Categorical (nominal) | Locale classification of the school (Urban, Partially Urban, Rural)                                                                                   |
| Management        | Categorical (nominal) | School management type (DepEd Managed)                                                                                                                |



|                                                               |                       |                                                                                                                                                                                                                |
|---------------------------------------------------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Level                                                         | Categorical (ordinal) | Grade offering of the school (Purely ES, ES and JHS, All Offering K to 12)                                                                                                                                     |
| n_test_takers                                                 | Numeric (discrete)    | Total number of learners who took the CRLA in the school                                                                                                                                                       |
| school_[band]_total /<br>_percent                             | Numeric (continuous)  | School-wide count and percentage of learners falling into each of the five CRLA reading bands (Low Emerging, High Emerging, Developing, Transitioning, At Grade Level), aggregated across grades and languages |
| g1_mt_[band]                                                  | Numeric (discrete)    | Grade 1 reading-band counts assessed in the Mother Tongue (MT)                                                                                                                                                 |
| g2_mt_[band] /<br>g2_fil_[band]                               | Numeric (discrete)    | Grade 2 reading-band counts assessed in Mother Tongue and in Filipino                                                                                                                                          |
| g3_mt_[band] /<br>g3_fil_[band] /<br>g3_eng_[band]            | Numeric (discrete)    | Grade 3 reading-band counts assessed in Mother Tongue, Filipino, and English                                                                                                                                   |
| g1_mt_total,<br>g2_mt_total,<br>g2_fil_total,<br>g3_mt_total, | Numeric (discrete)    | Total number of learners assessed per grade-language combination (denominator for computing band proportions)                                                                                                  |



|                               |  |  |
|-------------------------------|--|--|
| g3_fil_total,<br>g3_eng_total |  |  |
|-------------------------------|--|--|

Table 2. Summary of Dataset Attributes

Of these, the school's language attribute (mother tongue) is the main independent variable, while the grade- and language-specific literacy band variables, especially the Grade 3 Filipino (g3\_fil\_) and Grade 3 English (g3\_eng\_) reading bands are the dependent variables representing literacy performance. The region and Classification (urban/rural) and Level attributes are control and grouping variables used to explore whether the relationship between mother tongue and literacy performance is different for geographic and/or school type.

Data Preprocessing

1. Data Cleaning

The raw data was examined for missing data, duplicate data and inconsistent categorical labels. Any rows that are missing in the Ownership, Classification, and Level columns were flagged; Ownership is the same for all rows (Public) and was not used in the modeling, but was kept for documentation purposes. The schools that do not have test takers (n\_test\_takers = 0) and the schools that have all zero counts for the literacy band (for all grades) were identified and excluded as they did not have usable information on literacy outcomes. The text areas (such as language, Municipality, Barangay) were standardized in terms of casing and spacing to avoid creation of duplicate category labels due to lack of consistency in text formatting.

2. Handling Low-Frequency Categories

Because several mother-tongue categories (e.g., Sambal, Surigaonon, Southern Sorsoganon, Chavacano, Bahasa Sug, Yakan) have very few observations, these low-frequency languages were grouped into an “Other Minority Languages” category. This step reduced sparsity and prevent unstable estimates in statistical tests and models caused by categories with fewer than 10 observations, while preserving the five dominant mother-tongue groups (Tagalog,



Waray, Sinugbuanong Binisaya, Ilocano, and Pangasinan/Kapampangan) as distinct analytical groups.

### **3. Data Transformation and Encoding**

Nominal categorical variables (Region, Language, Classification, Level) were one-hot encoded and were kept as factor/grouping variables for statistical tests like Pearson Correlation. Min-max scaling and z-score normalizations were used to normalize/scale numeric rate variables (percentages and derived indices). This is an important step before scaling all the features using Hierarchical Clustering to prevent the larger numbers dominating the models.

### **4. Data Splitting**

The cleaned and engineered data were split into a training set and a testing set in an 80:20 ratio, with the 80% allocated to the training set, and the 20% to the testing set, with the proportions of each region and group of mother-tongues maintained in both sets.

## **Modeling**

### **1. Descriptive and Exploratory Analysis**

Descriptive statistics (mean, median and standard deviations of at-grade-level rates) were calculated by mother tongue and region to describe the baseline literacy patterns before conducting inferential testing. Visible patterns and outliers were identified through exploratory data visualizations, such as grouped bar charts of at-grade-level rates by language, boxplots of literacy gap indices by region..

### **2. Unsupervised Learning (Hierarchical Clustering)**

Hierarchical clustering were used to find natural groupings among the schools either for PCA reduced features or for the normalized literacy rates. This method created a tree-like diagram (dendrogram) to classify schools into different "performance profiles" without relying on any pre-existing labels.

### **3. Predictive Modeling (Logistic Regression)**

Both binary and Multi-Class Classification Logistic Regression models were utilized by the team to examine the effect that region and mother tongue have on a school's success in literacy. The binary classification model identifies a school as "High Performing" if their



overall\_mps is equal to or above a pre-set median or 75th percentile mark-the policy standard practice. In the multi-class classification model, performance was sorted into 3 groups with reasonable divisions using overall\_mps. These are classified into Proficient ( 75%), Nearly Proficient (50-74%), Not Proficient (< 50%). These performance classification thresholds are based on the policy standard practices referenced by the Second Congressional Commission on Education (EDCOM 2, 2026).

These binary or multi-class classification models serve as crucial models because they move from predicting a continuous output of overall\_mps to discrete policy outputs that educational policymakers understand and apply readily. The model employed the one-hot encoded categorical predictors (Region, Urban/Rural classification, Mother Tongue) to predict the likelihood of a school reaching this high literacy level. It computes odds ratios in order to determine just how much the Odds are multiplied/divided to observe the effect of the various regions (urban/rural), Mother Tongues on having a majority of positive literacy outcome while all the other factors remain constant. Accuracy and reliability were confirmed using 20% test data saved prior to modelling.

The binary models highlighted potentially replicable successful practices, and the multi-class classification models provide greater specificity about varying degrees of performance, enabling more targeted support for schools in need of improvement. Eventually, the models accuracy and reliability are being tested using Accuracy, weighted ROC-AUC, confusion matrices, and comprehensive classification reports.



RESULTS AND DISCUSSION

This section details the study’s findings, following the same sequence that was laid out in the methodology. To clearly answer the core research questions, the results were broken down into three organized parts: Research Question 1 which tackles the demographic and geographic profile of the 6,121 participating public elementary schools and the 647,681 learners they tested; Research Question 2 examines the early-grade reading proficiency across the five CRLA reading bands, which is separated by region and by school classification; Then, Research Question 3 explores the correlation between Mother Tongue (L1), Filipino (L2), and English (L3) proficiency at Grade 3. Followed by the correlation of Mother Tongue to Filipino and English proficiency and the underlying factors affecting the results. All tables and figures referenced below were generated from the finalized CRLA 2025-2026 dataset.

Demographic and Geographic Profile of Participating Schools

Region and Mother Tongue

| Region                          | No. of<br>Schools | % of<br>Schools | No. of<br>Learners | % of<br>Learners |
|---------------------------------|-------------------|-----------------|--------------------|------------------|
| Region VIII (Eastern Visayas)   | 2,024             | 33.07%          | 146,709            | 22.65%           |
| Region III (Central Luzon)      | 1,384             | 22.61%          | 272,607            | 42.09%           |
| Region I (Ilocos Region)        | 1,131             | 18.48%          | 112,796            | 17.42%           |
| Region IX (Zamboanga Peninsula) | 810               | 13.23%          | 70,091             | 10.82%           |
| CAR                             | 772               | 12.61%          | 45,478             | 7.02%            |

Table 3. Number of Participating Schools and Learners Tested, by Region

Table 3 shows how the participating schools and tested learners are spread across the 5 CRLA areas. Despite having the largest number of schools, which is 2, 024 with 33.07% of the total sample in Eastern Visayas (Region VIII), the numbers of learners are not the highest. On the other hand, although the number of schools is low in Central Luzon (Region III) (around 1,384



or 22.61% of the sample), the number of learners is high in the region (272, 607 or 42.09%). This reveals that, on average, schools in region III are much larger than those in the four regions, and as such, have a significant number of pupils and are likely to have a significant impact on the overall literacy rates of the region as compared to the number of schools. The Cordillera Administrative Region (CAR) is again smallest in terms of the number of schools (772 at 12.61% of the total) and the amount of learners tested (45,478 at 7.02% of the total) which represents smaller mountain school divisions.

| Mother Tongue         | No. of<br>Schools | % of<br>Schools | No. of<br>Learners | % of<br>Learners |
|-----------------------|-------------------|-----------------|--------------------|------------------|
| Tagalog               | 2,135             | 34.88%          | 336,387            | 51.94%           |
| Waray                 | 1,435             | 23.44%          | 99,875             | 15.42%           |
| Sinugbuanong Binisaya | 1,352             | 22.09%          | 112,527            | 17.37%           |
| Ilocano               | 940               | 15.36%          | 57,298             | 8.85%            |
| Pangasinan            | 157               | 2.56%           | 25,511             | 3.94%            |
| Kapampangan           | 83                | 1.36%           | 14,335             | 2.21%            |
| Sambal                | 7                 | 0.11%           | 810                | 0.13%            |
| Other                 | 4                 | 0.07%           | 116                | 0.02%            |
| Surigaonon            | 3                 | 0.05%           | 293                | 0.05%            |
| Southern Sorsoganon   | 2                 | 0.03%           | 97                 | 0.01%            |
| Bahasa Sug            | 1                 | 0.02%           | 123                | 0.02%            |
| Chavacano             | 1                 | 0.02%           | 241                | 0.04%            |
| Yakan                 | 1                 | 0.02%           | 68                 | 0.01%            |

Table 4. Number of Participating Schools and Learners Tested, by Mother Tongue

Looking at the data from Table 4 it shows that Tagalog is by far the most widely used medium of instruction in this sample. It is used in 2,135 schools (34.88%) and reported to have an impressive 336,387 learners (51.94% of all learners tested). The fact that Tagalog schools represent only about a third of all schools, reinforces earlier observations about campus sizes



where most Tagalog-medium schools are concentrated, especially in Region III. The rest of major languages are distributed Waray 1,435 schools (23.44%) and Sinugbuanong Binisaya 1,352 schools (22.09%) form the second and third most common mother tongues, while Ilocano accounts for 940 schools (15.36%). In contrast to the remaining mother tongue categories (Pangasinan, Kapampangan, Sambal, Surigaonon, Southern Sorsoganon, Chavacano, Bahasa Sug, Yakan and Other) only represent a fewer than 5% of schools, with some having less than ten schools nationwide. This confirms the data-preprocessing decision to group these low frequency languages into a single “Other Minority Languages” group for statistical modeling. Since categories such as Bahasa Sug, Chavacano and Yakan are represented only by one single school in the dataset which leads to highly unstable estimates that affect the inferential test.

| Region      | Ilocano | Kapa<br>mp. | Pangasi<br>nan | Sinug.<br>Binisaya | Tagalog | Waray | Other<br>* | Total |
|-------------|---------|-------------|----------------|--------------------|---------|-------|------------|-------|
| CAR         | 299     | 0           | 0              | 0                  | 469     | 0     | 4          | 772   |
| Region I    | 638     | 0           | 157            | 0                  | 336     | 0     | 0          | 1,131 |
| Region III  | 3       | 83          | 0              | 0                  | 1,291   | 0     | 7          | 1,384 |
| Region IX   | 0       | 0           | 0              | 790                | 16      | 0     | 4          | 810   |
| Region VIII | 0       | 0           | 0              | 562                | 23      | 1,435 | 4          | 2,024 |
| Total       | 940     | 83          | 157            | 1,352              | 2,135   | 1,435 | 19         | 6,121 |

*\*Others include Sambal, Surigaonon, Southern Sorsoganon, Bahasa Sug, Chavacano, and Yakan.*

**Table 5. Cross-Tabulation of Region by Mother Tongue (Number of Schools)**

As can be seen in Table 5, there is a definite regional concentration of language use, with none of the languages evenly distributed throughout the area. This highly localized setup is highly sensible considering DepEd Order No. 16, s. Deliberately allotted MTs to some school divisions in 2012. The data is in tune with this policy. For example, Kapampangan and Pangasinan are most likely to be found in Region III and Region I. Among the 1,435 schools



speaking Waray, all of them are in Region VIII, even the large linguistic blocks. It's a testament to how these linguistic walls have stayed pretty much intact throughout the years.

| Region                          | Dominant Mother Tongue | Schools Using It | No. of Schools |
|---------------------------------|------------------------|------------------|----------------|
| CAR                             | Tagalog                | 469              | 772            |
| Region I (Ilocos Region)        | Ilocano                | 638              | 1,131          |
| Region III (Central Luzon)      | Tagalog                | 1,291            | 1,384          |
| Region IX (Zamboanga Peninsula) | Sinugbuanong Binisaya  | 790              | 810            |
| Region VIII (Eastern Visayas)   | Waray                  | 1,435            | 2,024          |

Table 6. Dominant Mother Tongue per Region

Table 6 identifies the single dominant mother tongue in each region: Tagalog dominates in both CAR (469 of 772 schools) and Region III (1,291 of 1,384 schools), Ilocano dominates Region I (638 of 1,131 schools), Sinugbuanong Binisaya dominates Region IX (790 of 810 schools), and Waray dominates Region VIII (1,435 of 2,024 schools, i.e., all Waray-medium schools in the dataset are located in this one region). Because each region is functionally associated with one primary mother tongue, region and mother tongue are not fully independent grouping variables in this dataset; this needs to be kept in mind whenever regional and linguistic patterns move together in the analysis that follow.

School Classification (Urban, Partially Urban, Rural)

| Classification  | No. of Schools | % of Schools | No. of Learners | % of Learners |
|-----------------|----------------|--------------|-----------------|---------------|
| Partially Urban | 5010           | 81.8%        | 450,264         | 69.52%        |



| Classification | No. of<br>Schools | % of Schools | No. of<br>Learners | % of<br>Learners |
|----------------|-------------------|--------------|--------------------|------------------|
| Rural          | 618               | 10.1%        | 30,055             | 4.64%            |
| Urban          | 484               | 7.9%         | 148,367            | 22.91%           |
| Not Reported   | 9                 | 0.1%         | 687                | 2.93%            |

Table 7. Number of Participating Schools and Learners Tested, by School Classification

Table 7 shows that Partially Urban schools make up the overwhelming majority of the sample (5010 schools, 81.8%), while Rural (628 schools, 10.1%) and Urban (484 schools, 7.9%) schools are comparatively rare; 9 schools (0.1%) had no recorded classification and were retained under a "Not Reported" label rather than dropped, per the data-cleaning procedure described in the Methodology. Notably, Urban schools account for a disproportionately large 22.91% of all learners tested despite representing only 7.65% of schools, again pointing to larger average school sizes in urbanized areas, consistent with the pattern already observed for Region III and for Tagalog-medium schools.

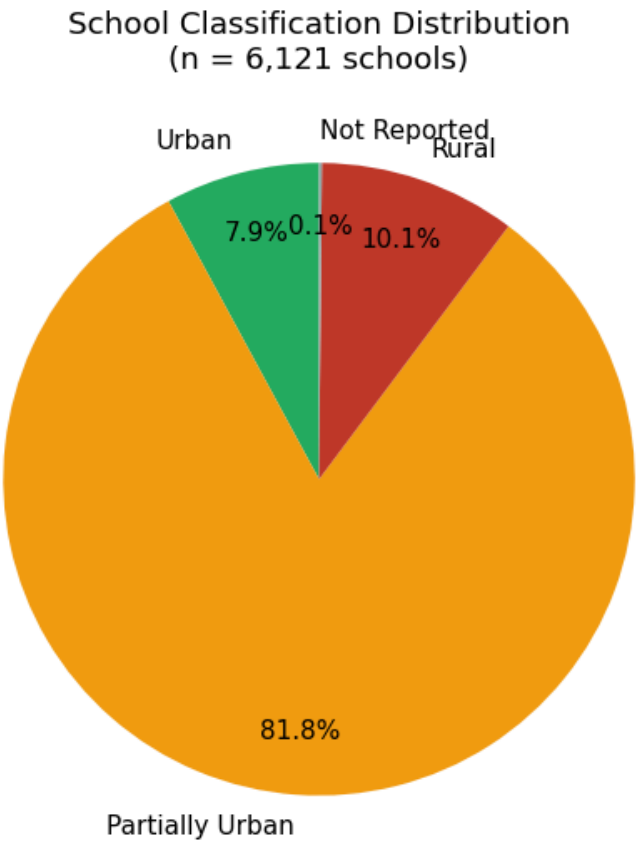




Figure 1. School Classification Distribution (n = 6,121 schools)

It's clear from **Figure 1** that there's a disparity between the categories. More than 4/5 of the schools are Partially Urban (81.8%), effectively upending the rest of the chart. The remaining data is divided into much smaller chunks: Rural schools account for 10.1%, Urban schools for 7.9% and just 0.1% for those that did not report their status.

This skewed distribution is an important factor to keep in mind when looking at any future comparisons of student performance between the partially urban schools and the others, since the size of the partially urban schools sample is much larger than the others. Rather, the results suggest that there is a large group of people at the majority end and a few smaller groups at the minority end, as is important to note when considering the differences in performance to be discussed later in this section.

| Region      | Urban  | Partially Urban | Rural  | Not Reported |
|-------------|--------|-----------------|--------|--------------|
| CAR         | 5.44%  | 48.83%          | 42.49% | 3.24%        |
| Region I    | 12.02% | 81.52%          | 3.89%  | 2.56%        |
| Region III  | 17.63% | 74.35%          | 4.34%  | 3.68%        |
| Region IX   | 0.00%  | 86.91%          | 8.89%  | 4.20%        |
| Region VIII | 2.27%  | 90.76%          | 4.59%  | 2.37%        |

Table 8. School Classification as a Percentage of Schools within Each Region

The extent of urbanization of the schools in each region is significantly different, when classified in this way (**Table 8**). By far the highest proportion of schools in the Rural categories is at CAR (42.49%) and this is likely related to the more sparsely populated and mountainous divisions of this region. Almost all of the schools in Region VIII (90.76% Partially Urban) and Region IX (86.91% Partially Urban, 0.00% Urban) are Partially Urban schools. The high economic development of Region III is also reflected in its high literacy outcomes discussed in the next section, as the most economically developed and industrialized region in the country, per



its sample. As the most economically developed and industrialized region in the country, per its sample, Region III has one of the fastest regional economic growth rates in the country, at 6.5% (Philippine Statistics Authority Regional Statistical Services Office III [PSA RSSO III], 2025), which is reflected in its high literacy outcomes, as discussed in the next section.

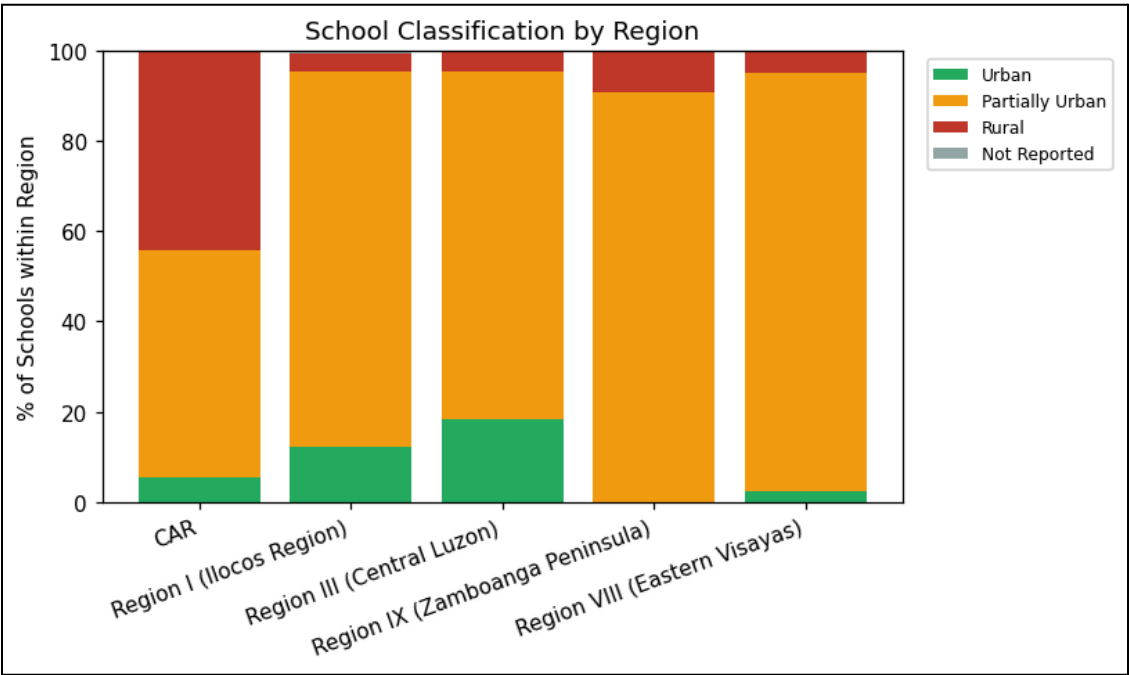


Figure 2. School Classification Mix within Each Region

The contrast between the regions in **Figure 2**, particularly at CAR. This is quite different from the other areas, where CAR's bar is a huge red rural section that dominates almost the entire bar with a score of 44%. By contrast, the rural blocks of any other region appear in the narrow margins at the top of each of their corresponding columns.

There are also some obvious regional variations with regard to the distribution of urban and partially urban schools. The middle part of each bar is almost exclusively urban, but the green urban portion at the bottom of each bar is not. The region with the biggest urban footprint is Region III, with nearly 20%, and Region IX is completely devoid of urban presence, with their green block completely invisible. This variation gives a very erratic appearance to the chart; the thickness of the urban and rural bands is constantly changing, actualizing the very peculiar structure of each region.

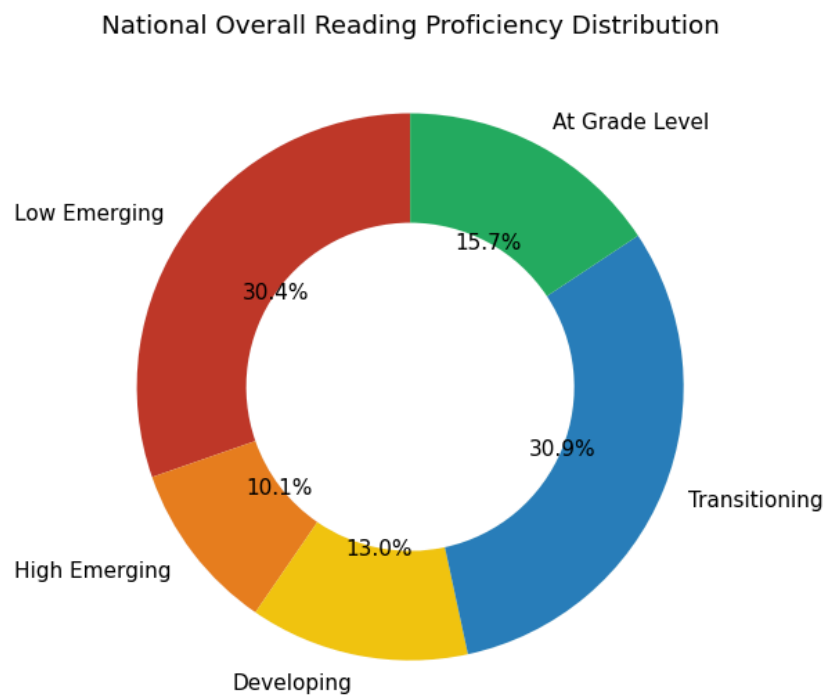


Status of Early-Grade Literacy per Region and Classification

| Reading Proficiency Level | % of Learners |
|---------------------------|---------------|
| Low Emerging              | 30.35%        |
| High Emerging             | 10.07%        |
| Developing                | 12.97%        |
| Transitioning             | 30.93%        |
| At Grade Level            | 15.68%        |

Table 9. National Overall Reading Proficiency Distribution

Table 9 presents the national baseline against which all regional and classification-level comparisons are made. Across all 647,681 learners tested, roughly three in ten (30.35%) remain at the Low Emerging level the lowest of the five CRLA reading bands while only 15.68%, or fewer than one in six learners, have reached the At Grade Level standard. Combining the two lowest bands (Low Emerging and High Emerging) as an indicator of literacy difficulty, 40.42% of all learners nationwide can be described as struggling readers, while 30.93% are Transitioning and 12.97% are Developing. This national picture is consistent with the learning crisis documented by EDCOM 2 (2024, 2026) and lends weight to the significance of examining what school- and language-level factors are associated with better or worse outcomes.



*Figure 3. National Overall Reading Proficiency Distribution*

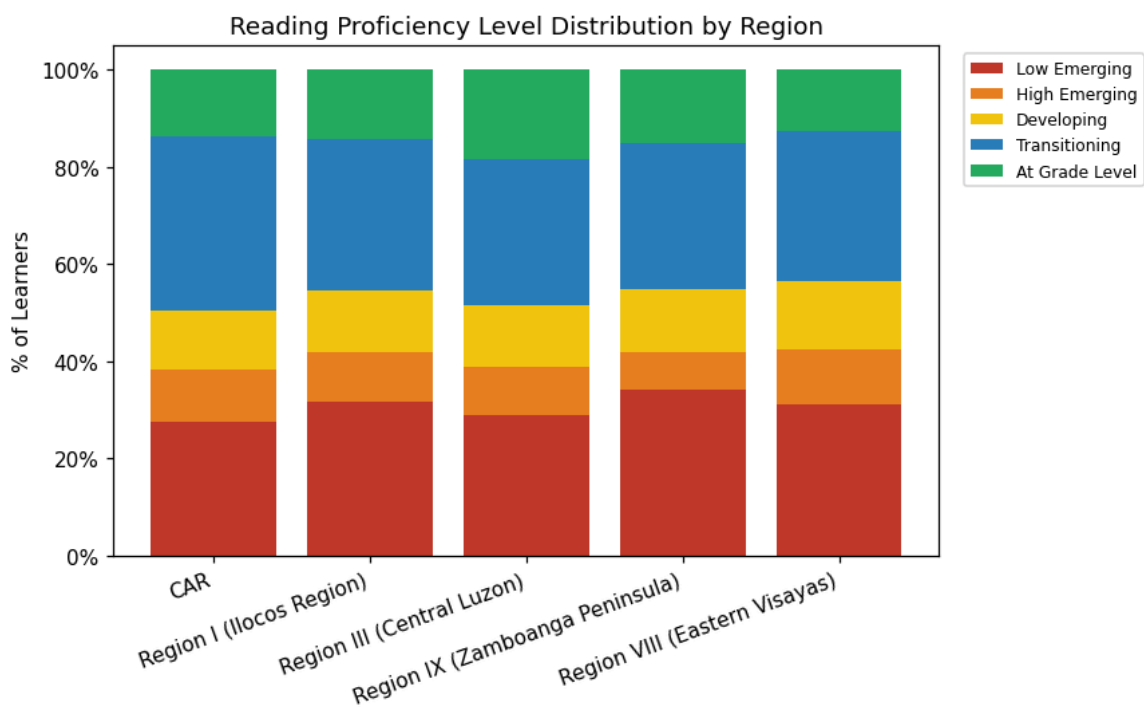
**Figure 3** renders the national distribution in **Table 9** as a donut chart, and the visual arrangement highlights how close together the two extreme bands are in size despite representing opposite ends of the proficiency scale: the Low Emerging wedge (30.35%) and the Transitioning wedge (30.93%) are nearly identical in width, together forming roughly two-thirds of the circle, while the At Grade Level wedge the actual policy target is visibly the second-smallest slice on the chart, only slightly larger than High Emerging. The chart makes it visually clear that the "middle" of the reading-proficiency journey (Transitioning) and the "bottom" (Low Emerging) currently claim far more of the pie than the intended end point (At Grade Level) does.



| Region      | Low Emerging | High Emerging | Developing | Transitioning | At Grade Level |
|-------------|--------------|---------------|------------|---------------|----------------|
| CAR         | 27.60%       | 10.72%        | 12.04%     | 35.85%        | 13.79%         |
| Region I    | 31.58%       | 10.18%        | 12.88%     | 31.16%        | 14.20%         |
| Region III  | 28.85%       | 9.91%         | 12.63%     | 30.21%        | 18.40%         |
| Region IX   | 34.19%       | 7.61%         | 12.90%     | 30.33%        | 14.97%         |
| Region VIII | 31.22%       | 11.26%        | 14.00%     | 30.84%        | 12.69%         |

Table 10. Reading Proficiency Level Distribution by Region (% of Learners)

Table 10 breaks this down by region. Region III (Central Luzon) obtained the highest percentage of At Grade Level (18.40%) among all the five regions. The highest Low Emerging (34.19%) is experienced by Region IX (Zamboanga Peninsula) while the lowest is recorded by CAR (27.60%). On the other side of the spectrum, though, it is Region VIII (Eastern Visayas) that has the lowest At Grade Level (12.69%) share of the region, while Region VIII's Low Emerging rate (31.22%) is somewhat lower than Region IX's (31.26%). That is, Region IX's problem is located at the bottom of the distribution (a lot of learners are at Low Emerging or even at the very bottom), while Region VIII's problem is located at the top (few learners are at full grade-level mastery, and fewer are at the very bottom); both are different forms of underperformance, and would be masked by a single summary number.



**Figure 4. Reading Proficiency Level Distribution by Region**

**Figure 4** presents the regional proficiency bands as stacked, color-coded bars, and comparing the bars side by side visually confirms the two distinct underperformance patterns described above. The Region IX bar shows a visibly taller bottom (Low Emerging) segment than any of the other four bars, while its top (At Grade Level) segment is only moderately sized. The Region VIII bar tells the opposite visual story: its bottom segment is only moderately tall, but its top segment is the thinnest sliver of At Grade Level color among all five bars. The Region III bar, by contrast, stands out for having the thickest top segment of the group, visually confirming its lead in At Grade Level attainment even though its overall bar height (like all the others) sums to 100%.



| Classification  | Low Emerging | High Emerging | Developing | Transitioning | At Grade Level |
|-----------------|--------------|---------------|------------|---------------|----------------|
| Urban           | 28.70%       | 10.84%        | 12.24%     | 30.95%        | 17.27%         |
| Partially Urban | 30.85%       | 9.78%         | 13.22%     | 30.80%        | 15.35%         |
| Rural           | 29.46%       | 10.03%        | 13.64%     | 33.44%        | 13.44%         |
| Not Reported    | 32.84%       | 11.08%        | 11.71%     | 29.79%        | 14.58%         |

Table 11. Reading Proficiency Level Distribution by School Classification (% of Learners)

Table 11 shows the same breakdown by school classification. Urban schools post the best outcomes overall, with both the highest At Grade Level share (17.27%) and a comparatively low Low Emerging share (28.70%). Schools with no recorded classification ("Not Reported") have the highest Low Emerging share of any group (32.84%), which may partly reflect data-quality issues associated with these unclassified schools rather than a true classification effect. Rural schools have the lowest At Grade Level share among the three classified groups (13.44%), yet consistent with the pattern noted in the Methodology's rationale for retaining incomplete classification data Rural schools are not the weakest group on the Low Emerging measure (29.46%, lower than both Partially Urban and Not Reported); instead, Rural learners are disproportionately concentrated in the middle Transitioning band (33.44%, the highest of any group), suggesting that Rural schools succeed at moving learners out of the earliest reading stage but then struggle to carry them the rest of the way to full grade-level mastery, possibly due to more limited access to reading materials and reading-intervention resources in later grades.



| Region      | Classification | Low Emerging | At Grade Level           |
|-------------|----------------|--------------|--------------------------|
| Region III  | Urban          | 26.65%       | 19.39% (highest overall) |
| CAR         | Urban          | 29.23%       | 16.11%                   |
| Region I    | Rural          | 24.15%       | 15.71%                   |
| Region IX   | Rural          | 37.35%       | 15.81%                   |
| Region VIII | Rural          | 36.53%       | 9.29%                    |
| Region VIII | Urban          | 33.79%       | 7.82% (lowest overall)   |

Table 12. Reading Proficiency Level Distribution by Region and Classification

The interactive patterns that emerge from the combination of region and classification based on **Table 12**, are not apparent in either dimension alone. The strongest At Grade Level rate in the dataset is in Region III's urban schools (19.39%), which continues to make Region III the strongest region in the entire dataset. Strikingly, however, Urban schools in Region VIII post one of the worst At Grade Level rates of any subgroup (7.82%) lower even than Rural schools within the same region (9.29%). It shows that urban classification, alone, does not ensure improved literacy outcomes, rather: urban location interacts with, and may negate, the assumed benefit of urban classification. This is directly relevant to research question 4 in the original study design, which inquires if the relationship between region and classification is uniform, and suggests that it is not.

There needs to be a halt to the single-minded assumption that 'urban' status equates with a single benefit. The same trend was described in recent commentary on this same CRLA cycle in the following cities, which are all very urbanized cities in Metro Manila: several of these cities in Metro Manila have some of the highest percentages of students who are "struggling" readers in their early grades despite having the advantage of resources (Philippine Star, 2026). In some areas, urban schools have been even worse at meeting grade-level standards than rural schools. One-size-fits-all funding formulas simply fail to capture this, and frequently leave communities where it is most needed with funding resources while ignoring communities where the need is real.



Correlation Between Mother Tongue, Filipino, and English Proficiency

In the CRLA dataset, the grade level is the only grade level where all three languages (mother tongue, Filipino, and English) are measured in the same grade level cohort of students, which makes it the logical grade level for testing the major hypothesis of this study, cross-linguistic transfer. Table 12 shows the Pearson correlation coefficient between each language pair for the 5,778 schools where Grade 3 test takers participated in all three languages. Mother Tongue and Filipino have the highest and most significant correlation ( $r = .9162$ ) followed by Filipino and English ( $r = .8325$ ), and last for the lowest correlation but most significant ( $r = .8074$ ). The null hypothesis of no significant correlation between Mother Tongue proficiency and Filipino/English proficiency is rejected in favor of the alternative hypothesis since all three correlations are strong and significant. This finding confirms the DepEd rationale for MTB-MLE, which is in line with Cummins's (1979) Linguistic Interdependence Hypothesis, which stated that there is a direct relationship between the linguistic ability of students in their mother tongue and their ability to read in Filipino and English. It further replicates the size of the association reported by Abrigo et al. (2025), who analyzed the PIDS data set and came up with similar conclusions that linguistic mismatch creates a significant "tax on learning. It also corroborates the strength of the association reported by Abrigo et al. (2025) and provides empirical evidence of the same underlying premise with a different, more recent data set.

| Language Pair                        | Pearson r | p-value  | Significant<br>( $p < .05$ ) |
|--------------------------------------|-----------|----------|------------------------------|
| L1 (Mother Tongue) vs. L2 (Filipino) | 0.9162    | $< .001$ | Yes                          |
| L1 (Mother Tongue) vs. L3 (English)  | 0.8074    | $< .001$ | Yes                          |
| L2 (Filipino) vs. L3 (English)       | 0.8325    | $< .001$ | Yes                          |

Table 13. Pearson Correlation Between Grade 3 Mother Tongue, Filipino, and English Proficiency Scores (N = 5,778 Schools)

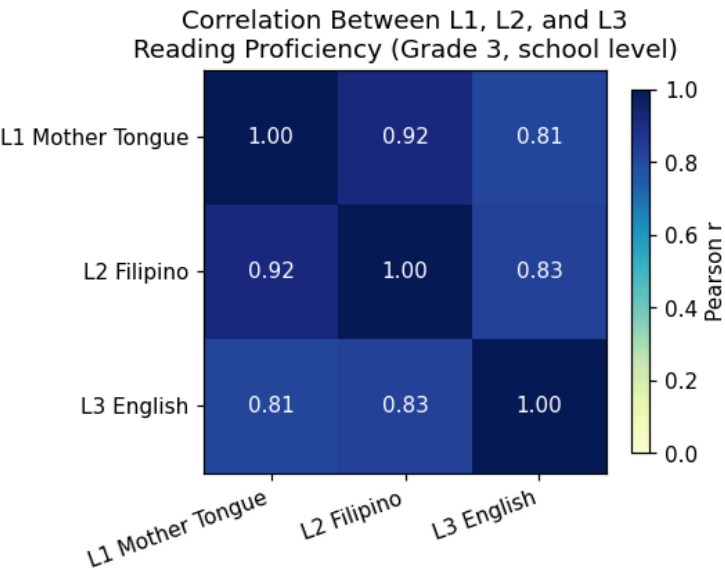


Figure 5. Correlation Heatmap Among Grade 3 L1, L2, and L3 Reading Proficiency

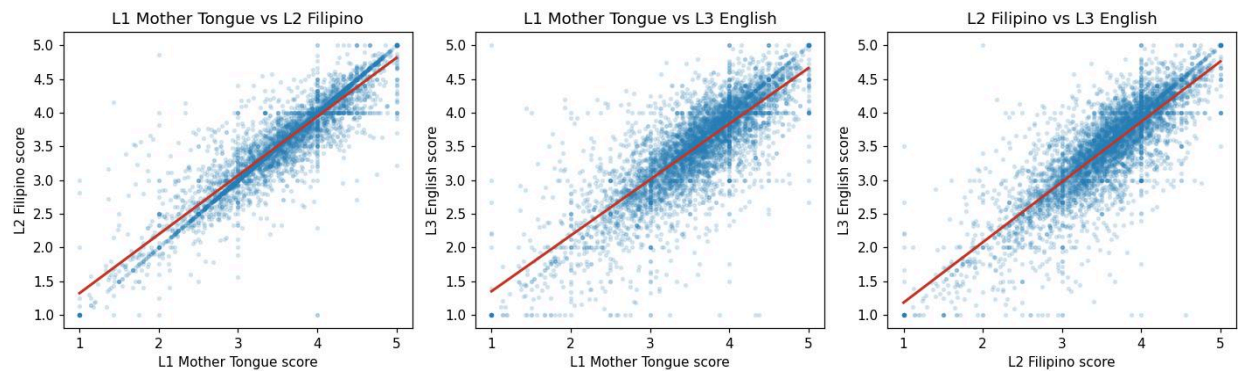


Figure 6. Scatter Plots with Fitted Trend Lines for Each Grade 3 Language Pair

The scatter plots in **Figure 6** visually confirm the linear, positive relationship underlying each coefficient, with the Mother Tongue–Filipino pair showing the tightest clustering around its trend line consistent with it having the highest correlation of the three. It is worth reiterating the study's stated limitation that these results describe correlation, not causation: a school whose learners read well across all three languages may do so because of the **L1-to-L2/L3** transfer mechanism the policy assumed, or because of a common underlying factor such as teacher quality, school resources, or the socioeconomic profile of the community that simultaneously drives proficiency in all three languages regardless of instructional medium. Household income, health and nutrition are not measured directly in the CRLA dataset used in this study and Kilag et al. (2024) point out that background factors such as household income, health and



nutrition—independently influence literacy outcomes, and therefore the high correlation they find should be read as evidence for, rather than against, the cross-linguistic transfer mechanism.

Correlations between a student's mother tongue and his/her English and Filipino scores are strong, and are consistent with recent studies on translanguaging in early literacy learning (Abrigo et al., 2025). The idea is that if a child has a strong foundation in his or her first language, the second or third language can catch up. The concept is that if there is a good reading base in the first language, the second or third language can catch up.

However, when this met the real world, the rollout of mother tongue education on a national scale actually had a weak (but noticeable) negative correlation with scores for English and Filipino (Igarashi et al., 2024). This was not because the language was faulty; in particular, it wasn't because the linguistic theory was flawed. It was caused by a lack of qualified teachers, delayed teaching materials, and classrooms where multiple local languages were used, among other factors (Igarashi et al., 2024). As there is a correlation between language skills across languages, language policies cannot simply be reviewed based on the quality of their position papers; they need to be sustained through funding, teacher training and effective support in the classroom (Kilag et al., 2024).

### **Relationship Between L1 Proficiency and L2/L3 Proficiency Across Regions, Classifications, and Language Groups**

To identify if the relationship between Mother tongue (L1) and Filipino/English language proficiency is consistent in different school settings, the researchers made use of Pearson correlation for each subgroup of learners (by region, school classification, and by mother tongue) in two language combinations (Mother tongue to Filipino/English).

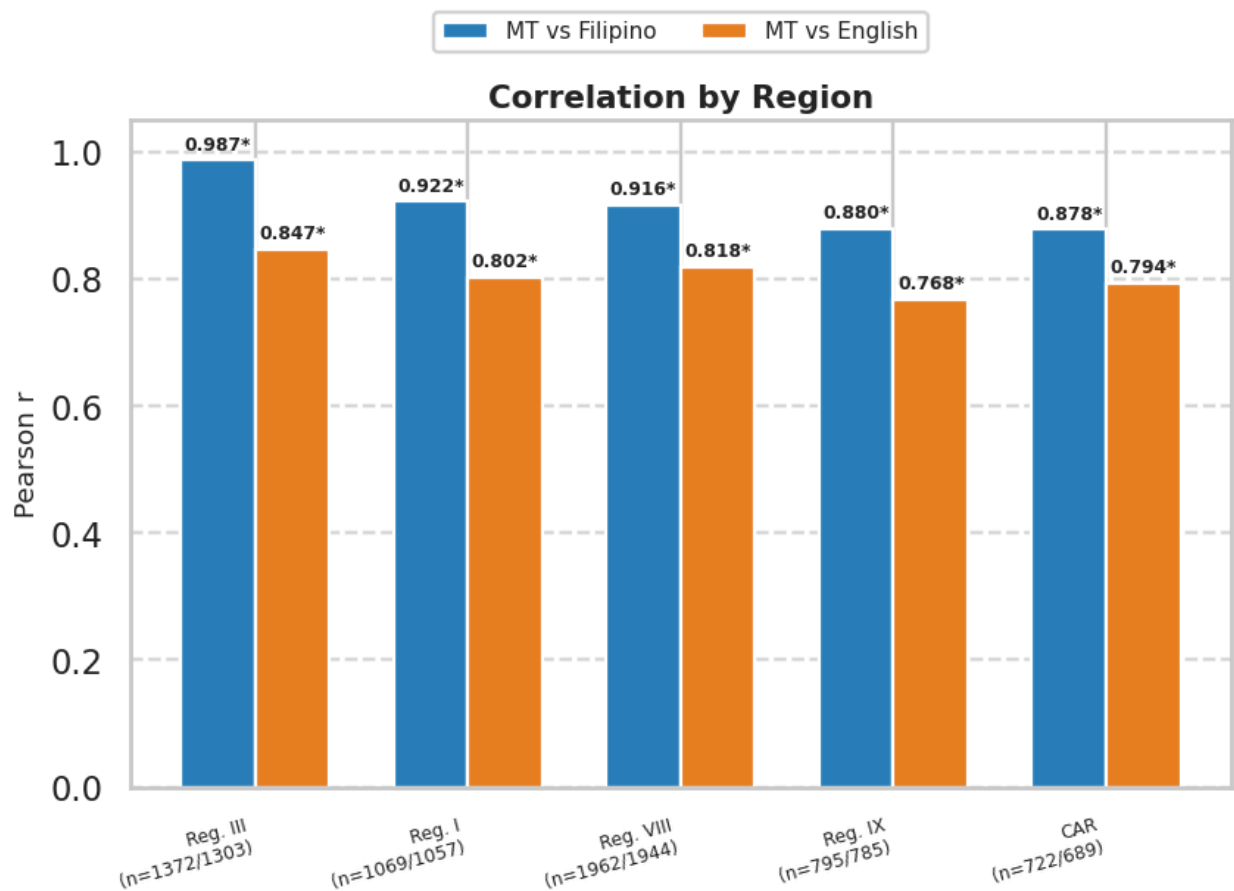


Figure 10. Correlation of L1–L2 and L1–L3 Proficiency, by Region

The high correlations in Region III (L1-L2:  $r = .987$ ; L1-L3:  $r = .847$ ) may indicate good MTB-MLE implementation within the region, more resources, and more homogeneity in language whereas the region mostly uses Tagalog, the base for Filipino. Near perfect L1-L2 correlation means that for Central Luzon, MTB and Filipino teaching are virtually testing the same construct of literacy.

CAR (L1-L2:  $r = .878$ ; L1-L3:  $r = .794$ ) and Region IX (L1-L2:  $r = .880$ ; L1-L3:  $r = .768$ ) have the smallest coefficients. This could be due to the higher linguistic diversity found within the region, which could lead to situations where the School’s appointed "Mother Tongue" may not be the actual students' home languages. Additionally, Region III displays the largest difference between L1-L2 and L1-L3 (0.140), while CAR shows the smallest (0.084). Showing that strong L1 knowledge is more transferable to Filipino than to English, possibly due to linguistic similarity.

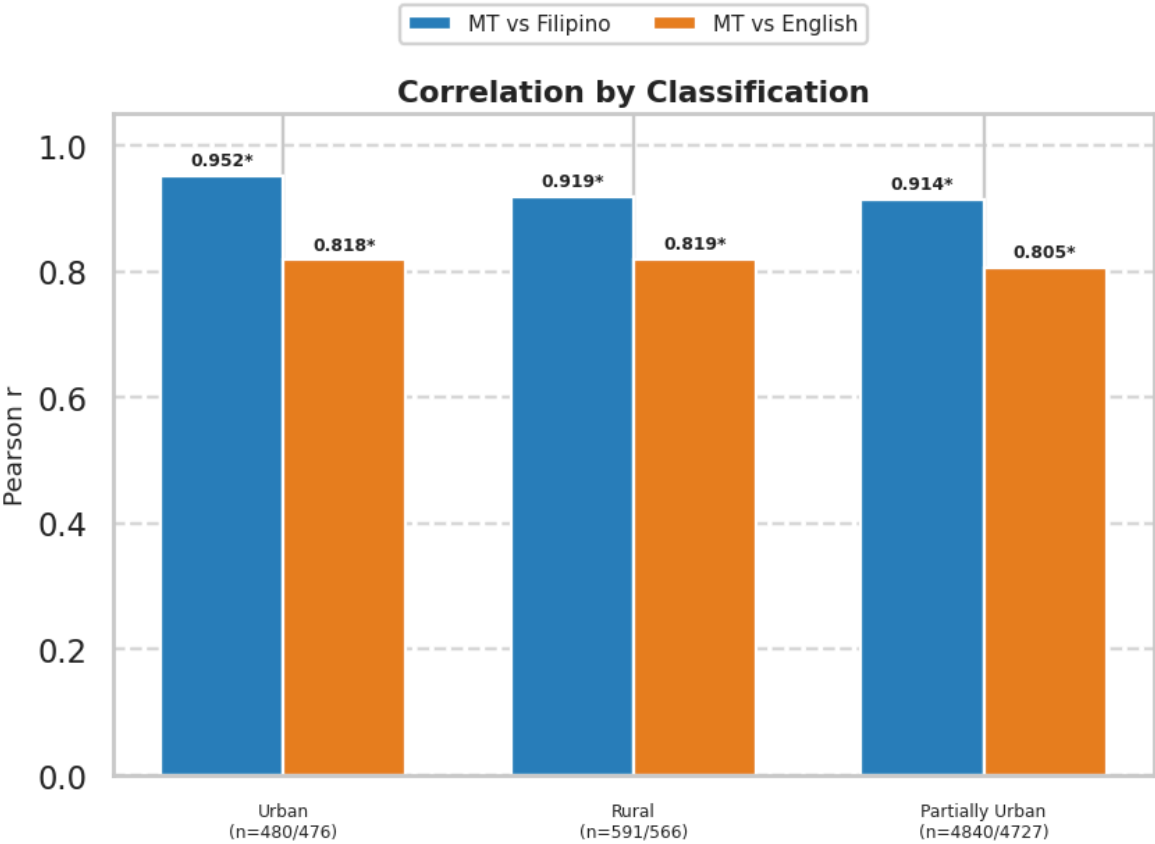


Figure 11. Correlation of L1–L2 and L1–L3 Proficiency, by Classification

Correlations indicate little variation by classification. The values for L1-L3 (0.818 for Urban, 0.819 for Rural and 0.805 for Partially Urban) closely match each other, showing that cross-linguistic transfer is not related to the level of the school resources. While the amount of resources in school could affect the literacy outcome, it does not affect the process or system of moving from L1 to L2/L3 (Kilag et al., 2024). The modest L1–L2 advantage for Urban (0.952 vs. 0.919/0.914) could be due to the more uniform way in which the Filipino language is taught, or because of more reading materials available.

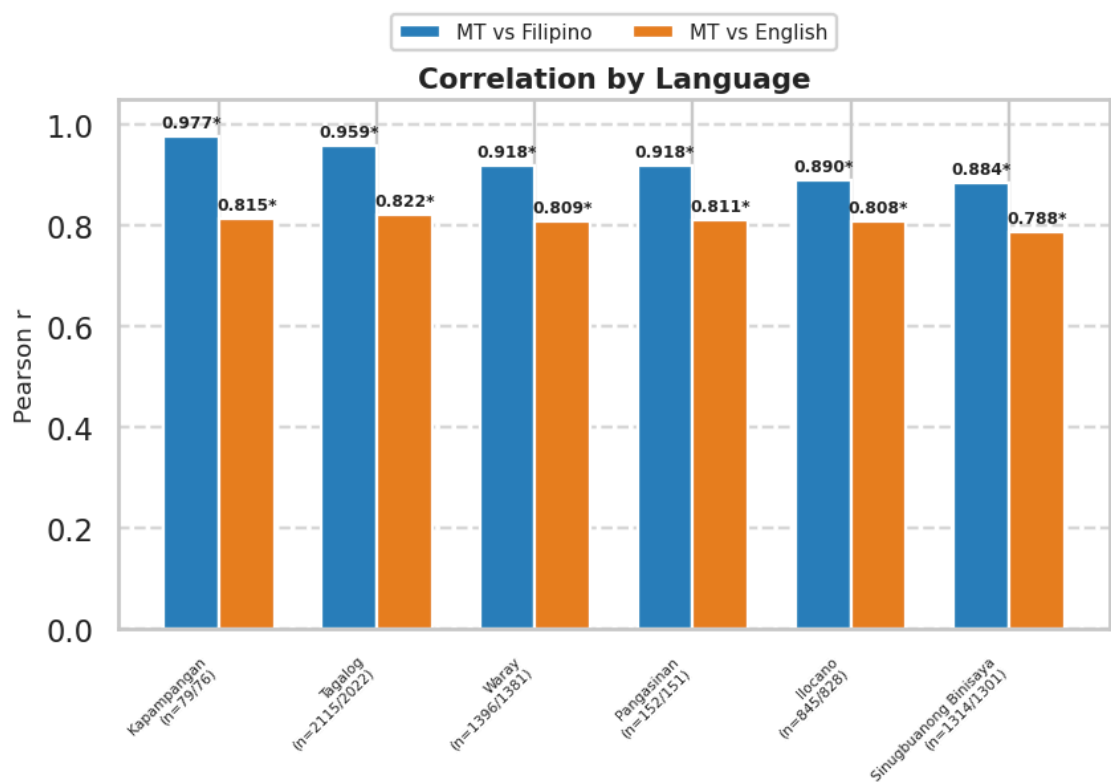


Figure 12. Correlation of L1–L2 and L1–L3 Proficiency, by Language

The languages which are linguistically more similar to Filipino tend to have higher correlations. The highest correlation value of 0.977 is Kapampangan and Filipino and followed by Tagalog, having a correlation value of 0.959 are the languages with the highest L1-L2 correlation. This shouldn't come as a shock because Filipinos are heavily influenced by Tagalog. Languages that are far more away from Filipino have weaker relationships in comparison. These are observed for Sinugbuanong Binisaya (0.884) and Ilocano (0.890) due to their very different grammar and vocabulary from Filipino. The disparity is even greater for English. In terms of L1–L3 learning correlation, the Sinugbuanong Binisaya had the lowest correlation, with 0.788, this suggests that students who speak the Visayan language may require additional support in linking their Visayan knowledge and language to English. The transfer is not as spontaneous with them as with the Tagalog speakers.

Overall, mother tongue is strongly connected to other languages throughout the region but this connection varies depending on: similarity to the language similarity, region, and classification. This correlation is the highest if the mother tongue is close to Filipino (Tagalog or



Kapampangan); and if the students come from language similar areas in the country (Central Luzon). This relationship is not strongly affected by school type (Urban or Rural) – it is related to language similarity rather than to resources. MTB-MLE works best when the mother tongue is related to Filipino. Teachers should make sure that for more distant languages, they will do their best to help students recognize the link between their language and the Filipino and English language.

**Distinct Performance Clusters Emerge from Grade 1–3 Reading Trajectories**

Schools were clustered using hierarchical clustering on a matrix consisting of the following six reading scores: Grade 1 Mother Tongue, Grade 2 Mother Tongue, Grade 2 Filipino, Grade 3 Mother Tongue, Grade 3 Filipino, and Grade 3 English. The cluster was defined by reading scores without any demographic variable such as region, language, etc. Schools with at least one non-zero test taker for each of the six test assessments were retained; 5,726 schools satisfied this criteria.

In order to find the best number of clusters, a silhouette analysis of different numbers of clusters from 2 to 8 was used. The silhouette score is a measure of similarity to one’s own cluster, and dissimilarity to the neighboring cluster; the higher the silhouette score the better the partitioning of data into clusters.

| Number of Clusters | Silhouette Score |
|--------------------|------------------|
| 2                  | 0.2807           |
| 3                  | 0.2021           |
| 4                  | 0.1514           |
| 5                  | 0.1434           |
| 6                  | 0.1228           |



|   |        |
|---|--------|
| 7 | 0.1295 |
| 8 | 0.1389 |

Table 16. Silhouette Score by Number of Clusters

As seen on **Table 16**, the silhouette scores show that the two-cluster solution has the highest score at 0.2807, a score substantially greater than the three cluster solution score of 0.2021. The two cluster solution would mean that the schools could most logically and clearly be separated into only two groups based on their reading progression. As the number of clusters increases, the score gradually reduces, showing no real benefit of further dividing the schools into additional groups. Therefore, the number of clusters was chosen to be two.

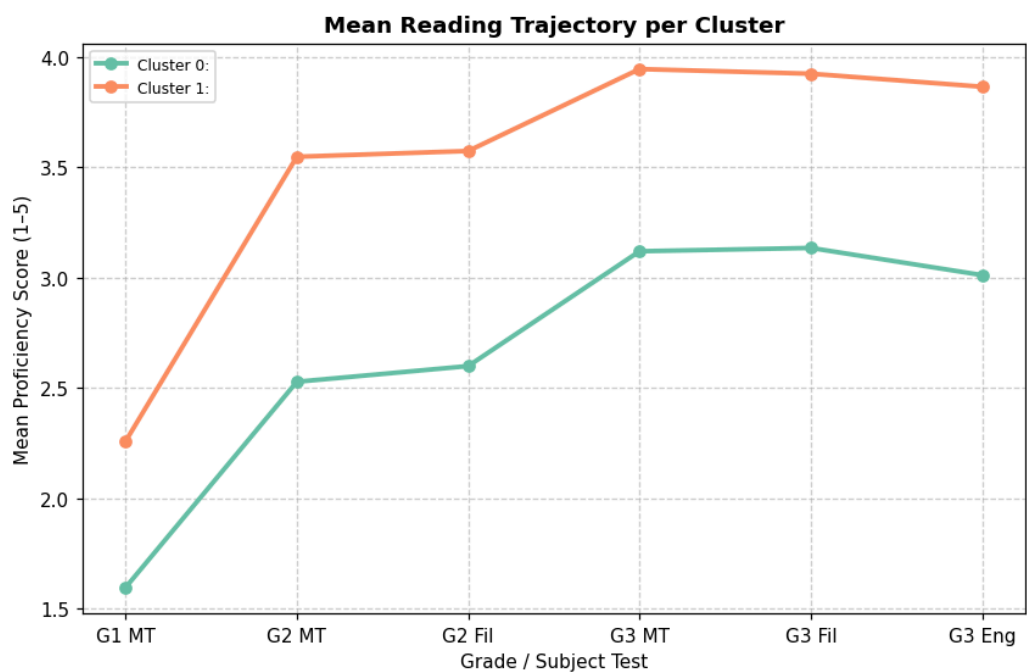


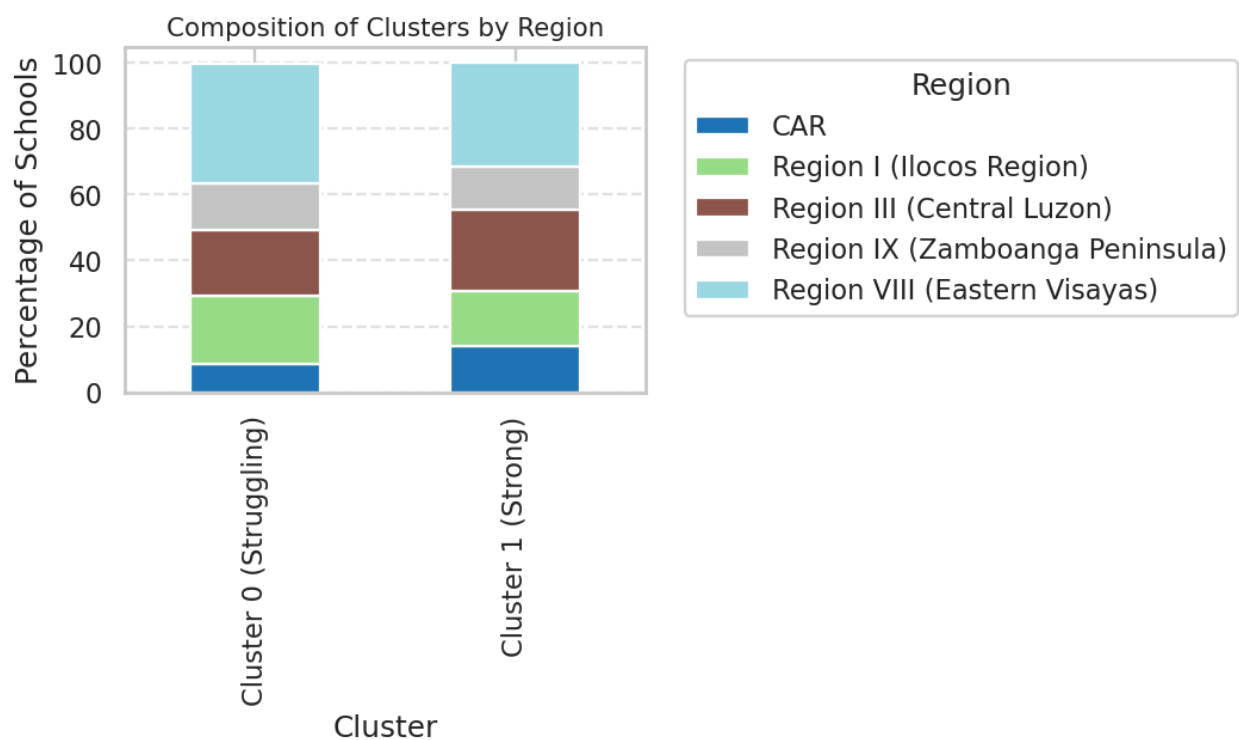
Figure 13. Mean Reading Trajectory per Cluster

According to Figure 13 the two clusters exhibit markedly distinct trends in reading skills acquisition from Grade 1 to Grade 3. Cluster 0 consists of 2538 schools and is entitled 'Struggling Schools'. These schools began with low Mother Tongue results at Grade 1 (1.55), well within the Low Emerging range. They proceed with a moderate, though slow, development through Grade 2 (Mother Tongue = 2.50, Filipino = 2.55) falling far behind the Strong



Performers. In spite of the jump from Grade 2 to Grade 3 (Mother Tongue = 3.15, Filipino = 3.15, English = 3.00) their overall acquisition of reading skills continues to fall significantly behind Cluster 1 throughout all subject areas. These clearly demonstrate an initial plateauing and gaps in their beginning reading with persistently below benchmark score. There may have been a lack of teachers with adequate preparation, less available learning tools or the difficulty in applying Mother Tongue Based-Multilingual Education in a class where different languages exist. There are 3,188 schools under Cluster 1, 55.7 percent of all schools identified as “Strong performers”.

The said cluster starts with a high initial of around 2.25 in Mother Tongue in Grade 1 in High Emerging. The Mother Tongue (3.55) and Filipino (3.60) for Grade 2 show phenomenal growth then sky-rocketed their Mother Tongue (3.95), Filipino (3.95), and English (3.90) by Grade 3 to near perfection. The curve of the graph is sharp, steady, upward progression at all measurement intervals, never going flat or downward. The success stories of MTB-MLE start here. They build the base in Grade 1 extremely strong, and do so with an exponential progression up into grades 2 and 3. They have sustained highest Mother Tongue mastery and consistently high performance in Filipino and English with translation to their other two languages being nearly perfect at the end of Grade 3.



**Figure 14. Composition of Clusters by Region**

Looking at the regional composition, the stacked bars highlight specific geographic strongholds and deficits. The light blue stack representing Region VIII (Eastern Visayas) visibly overtakes the struggling Cluster 0 bar, creating a sharp visual contrast against its smaller footprint in Cluster 1. This confirms Region VIII's outsized struggle with early-grade literacy.

In contrast, the brown segment for Region III (Central Luzon) takes up a visibly wider proportion of Cluster 1's bar, visually cementing it as the anchor of the "Strong Performers" cluster. Notably, the dark blue segment representing CAR (Cordillera Administrative Region) is visibly larger in the strong cluster's composition than in the struggling cluster. This visual evidence strongly reinforces the earlier finding that CAR is not a region of uniform struggle, but rather possesses a higher-than-expected concentration of effective schools that are successfully implementing MTB-MLE. Region I (light green) and Region IX (grey) show more balanced visual proportions, though both lean slightly toward the struggling cluster.

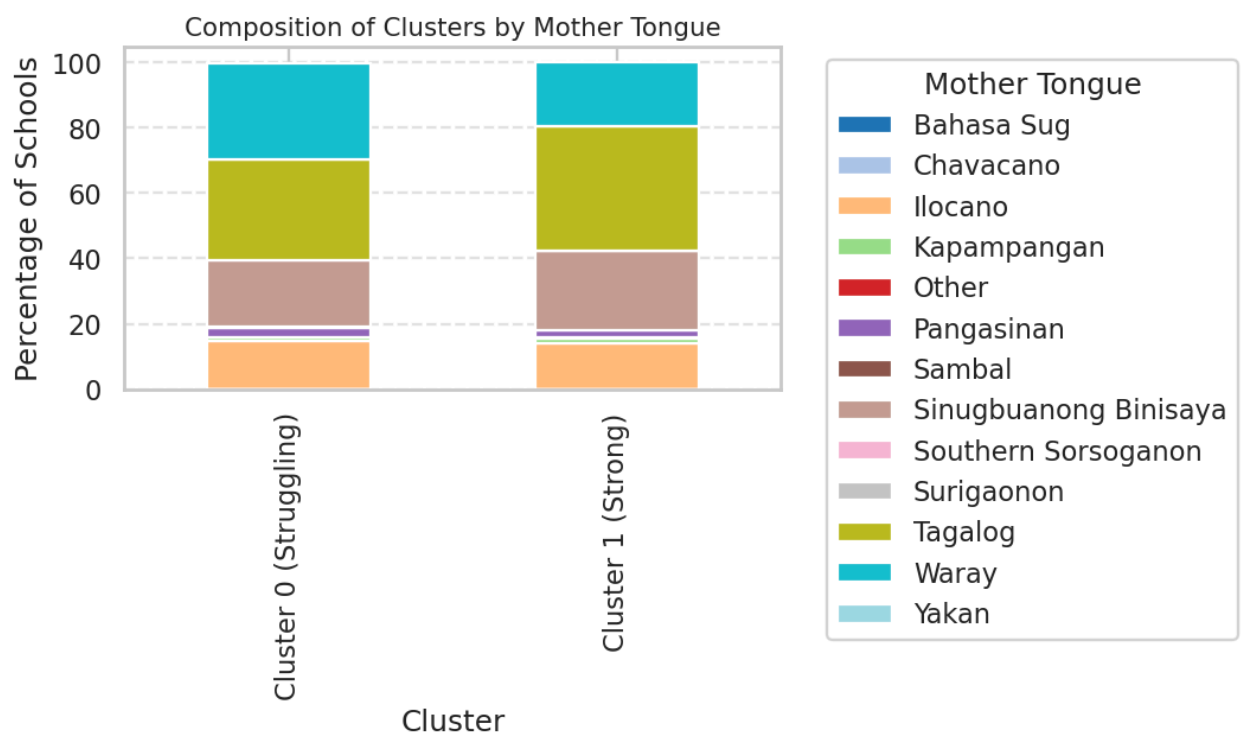


Figure 15. Composition of Clusters by Mother Tongue

A visual comparison of Mother Tongue composition of the schools shows the two clusters are very different visually. The widerteal/cyan section, which represents the Waray group of schools, of Cluster 0 (Struggling), which proves there are about 30% of Waray schools as weak performers as compared to the 19.5% that was seen in the strong performing cluster. However, the majority part, represented by a yellow-green slice which stands for schools that use Tagalog as mother tongue, takes a more apparent space on the bar chart of Cluster 1, which proves that Tagalog speaking schools form the strong core of all Strong Performers.

While the brown segment for Sinugbuanong Binisaya is present in both bars, it holds a slightly wider visual footprint in Cluster 1, once again challenging the previous assumption that Visayan languages inherently lead to struggle. The orange segment for Ilocano shows a relatively balanced visual width across both clusters, indicating it is a moderate presence in both the struggling and strong performances.

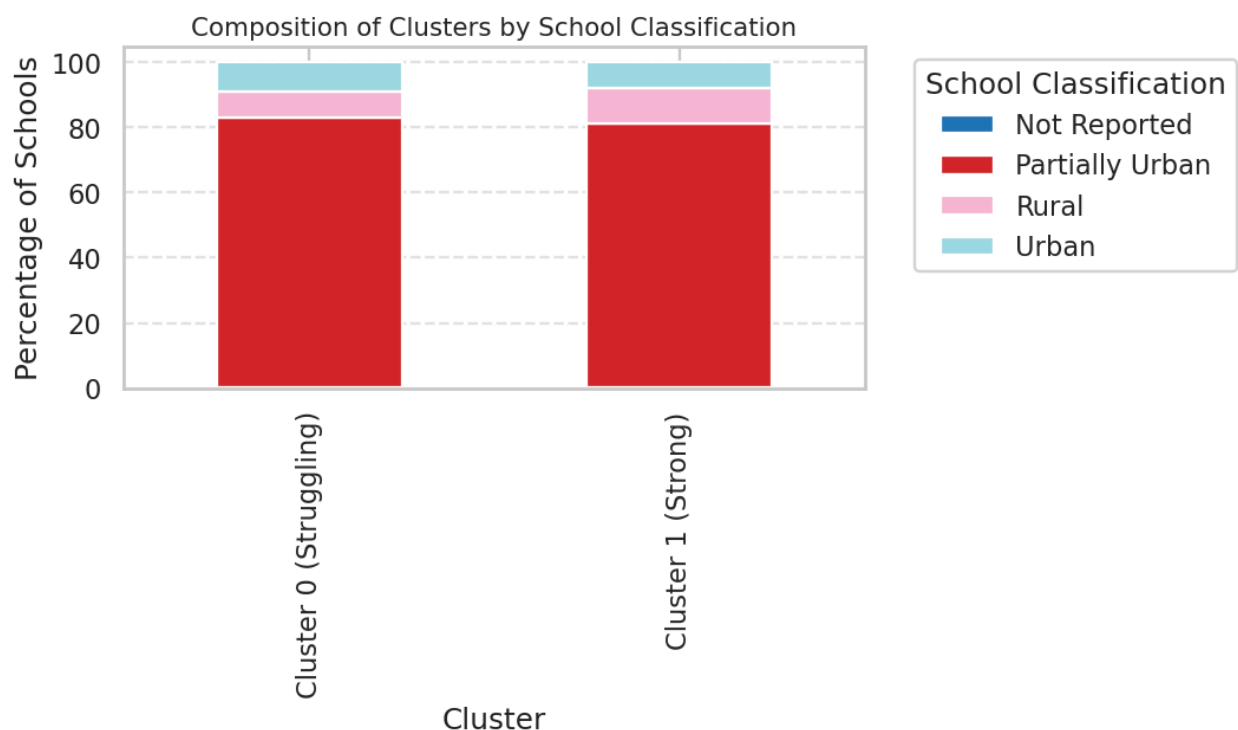


Figure 16. Composition of Clusters by School Classification

The two patterns are very similar, as indicated by the bar graphs. In both cases, there's an enormous, virtually identical red slice denoting Partially Urban schools; over 80% of schools in both patterns are from this group. The trajectory of reading progress is more influenced by the school context (urban vs rural) than by local teaching practices and instructional quality or local communities.

There are two natural groups of schools according to their pattern of reading growth. The larger group, 3,188 (55.7%), are the "Sustained Strong Performers" which begin well and make sustained, accelerated progress. The other group of schools (2,538 – 44.3%) are the “Struggling Schools” that start poorly, have poorer early grades performance and don't achieve the high-performance levels of the other schools. The strong group is dominated by Tagalog schools while the struggling group is dominated by Waray schools and Region VIII. However, success is not a foregone conclusion: School success is not determined by geography, linguistic distance or by categorising schools as Sinugbuanong Binisaya or CAR, as these settings are over-represented among strong performers, with rural schools showing little difference to urban schools. Administrative labels are not nearly as important as implementation quality. Policy will need to



provide extra support for failing Waray, Ilocano and Pangasinan schools as well as conduct intensive study of the high performing outliers in CAR and Sinugbuanong Binisaya to replicate 'best practice' in context.

### **Predictive Modeling: Determinants of School-Level Reading Success**

Mother tongue was aggregated to its six most represented categories (Tagalog, Waray, Sinugbuanong Binisaya, Ilocano, Pangasinan and Kapampangan, each with a sample of 30 schools and more) with Region and Classification combined and also coded as categorical predictors in a set of binary Logistic Regression models. This was done because the national At-Grade-Level rates are low (Table 8) and so the school was only considered to be "high-proficiency" if the percentage of Grade 3 students who tested at At Grade Level or higher was at least 20%, and "high-performing" if the percentage of students in its combined Low Emerging/High Emerging bands was at or above 30% of the test-takers, rather than an arbitrary 50% cut-off. The training data was class balanced, using random undersampling, for each model, and test data was a held-out 20% sample of the data; coefficients were exponentiated into odds ratios (OR), where  $OR > 1.0$  points toward a feature that increases the odds of achieving high proficiency and  $OR < 1.0$  indicates a feature that decreases the odds of achieving high proficiency, while the other predictors are held constant.

Four models were estimated to triangulate the Mother-Tongue effect from different perspectives: (1) a Mother-Tongue-only model that predicts high English performance without controlling for Region or Classification; (2) a full model that predicts high English performance (low struggling-reader) given a combination of Region, Classification and Mother-Tongue; (3) a full model predicting high Grade 3 English performance; and (4) a full model predicting high Grade 3 Filipino performance. Table 15 shows the classification accuracy for all four models.



| Model | Target                              | Predictors                              | N       | Accuracy | Precision | Recall | F1     |
|-------|-------------------------------------|-----------------------------------------|---------|----------|-----------|--------|--------|
|       | Outcome                             |                                         | Schools |          | (High)    | (High) | (High) |
| 1     | High English (≥20% At Grade Level)  | Mother Tongue only                      | 5,890   | 55.52%   | 0.5113    | 0.5456 | 0.5279 |
| 2     | High-Performing (≤30% struggling)   | Region + Classification + Mother Tongue | 6,093   | 61.44%   | 0.3435    | 0.4332 | 0.3832 |
| 3     | High English (≥20% At Grade Level)  | Region + Classification + Mother Tongue | 5,890   | 54.41%   | 0.5000    | 0.5829 | 0.5383 |
| 4     | High Filipino (≥20% At Grade Level) | Region + Classification + Mother Tongue | 6,028   | 58.62%   | 0.5615    | 0.3153 | 0.4038 |

Table 17. Logistic Regression Model Performance Summary (Held-Out Test Set)

The overall accuracy is quite limited for all four specifications, ranging from 54.41% to 61.44%. This is significantly higher than the majority-class floor for the more balanced English and Filipino targets (around 54–55%) and slightly higher than that for the struggling-reader target (around 72% of schools is the majority-class floor). No one of the models comes close to the sort of accuracy that would allow Mother Tongue, Region, and Classification alone to



accurately predict the level of achievement in any given school; what remains constant across the models is that the latter are statistically significant but quantitatively relatively weak predictors of school-level achievement, as noted below.

Isolating the Effect of Mother Tongue on High English Proficiency

To isolate the Mother Tongue effect from the confound of geography (since, as Table 4 and Table 5 showed, mother tongue and region are highly overlapping), Model 1 predicts high English proficiency using language alone, with no Region or Classification terms. Table 16 reports the resulting odds ratios and Figure 13 displays them graphically.

| Mother Tongue         | Coefficient | Odds Ratio |
|-----------------------|-------------|------------|
| Kapampangan           | 0.8242      | 2.2801     |
| Tagalog               | 0.2257      | 1.2532     |
| Ilocano               | 0.0277      | 1.0281     |
| Sinugbuanong Binisaya | −0.1517     | 0.8592     |
| Pangasinan            | −0.3147     | 0.7300     |
| Waray                 | −0.5437     | 0.5806     |

Table 18. Odds Ratios: Effect of Mother Tongue on High Grade 3 English Proficiency

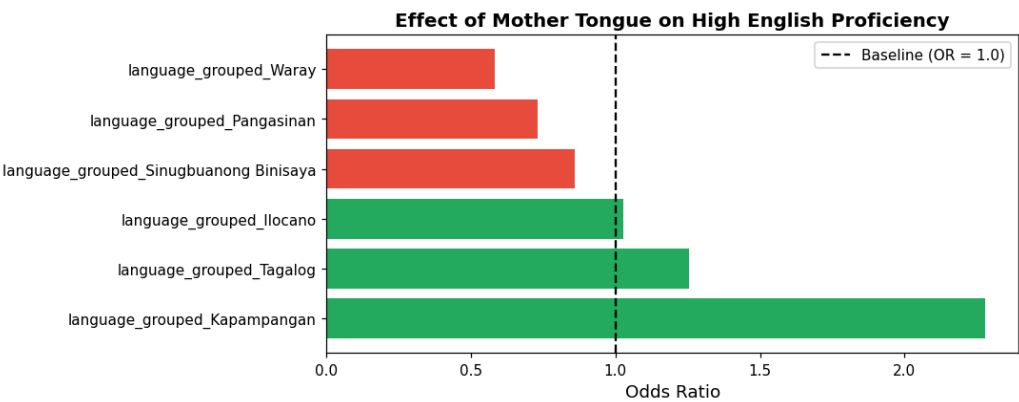


Figure 17. Effect of Mother Tongue on High English Proficiency (Model 1)

The odds of achieving high English proficiency are more than double in Kapampangan-medium schools compared to the model's reference level (OR = 2.2801), followed by Tagalog (OR = 1.2532) and Ilocano (OR = 1.0281) close to the neutral OR = 1.0 level. The schools with the lowest odds of being proficient in English are Waray (OR = 0.5806) and



Pangasinan (OR = 0.7300) schools. The Mother-Tongue-only result is challenging to interpret as it is not unexpected that the local economy in Region III (where Kapampangan and Tagalog schools are clustered) is stronger than that in the rest of the country, as Region III's regional growth is above average (PSA RSSO III, 2025) and this effect is explored further in Model 3.

**Determinants of High-Performing (Low Struggling-Reader) Schools**

Model 2 predicts whether a school keeps its combined Low Emerging and High Emerging ("struggling-reader") rate at or below 30%, using Region, Classification, and Mother Tongue jointly. Table 17 and Figure 14 report the strongest predictors.

| Feature                         | Coefficient | Odds Ratio |
|---------------------------------|-------------|------------|
| Sinugbuanong Binisaya           | 0.4172      | 1.5177     |
| CAR                             | 0.4049      | 1.4992     |
| Classification: Rural           | 0.2978      | 1.3469     |
| Region VIII (Eastern Visayas)   | 0.2219      | 1.2484     |
| Classification: Partially Urban | 0.0928      | 1.0973     |
| Kapampangan                     | 0.0855      | 1.0893     |
| Ilocano                         | 0.0830      | 1.0865     |
| Tagalog                         | 0.0222      | 1.0225     |
| Region I (Ilocos Region)        | -0.0410     | 0.9599     |
| Region III (Central Luzon)      | -0.1785     | 0.8365     |
| Pangasinan                      | -0.2844     | 0.7525     |
| Waray                           | -0.3913     | 0.6761     |
| Classification: Urban           | -0.4585     | 0.6322     |
| Region IX (Zamboanga Peninsula) | -0.4751     | 0.6218     |



Table 19. Strongest Predictors of High-Performing ( $\leq 30\%$  Struggling-Reader) Schools

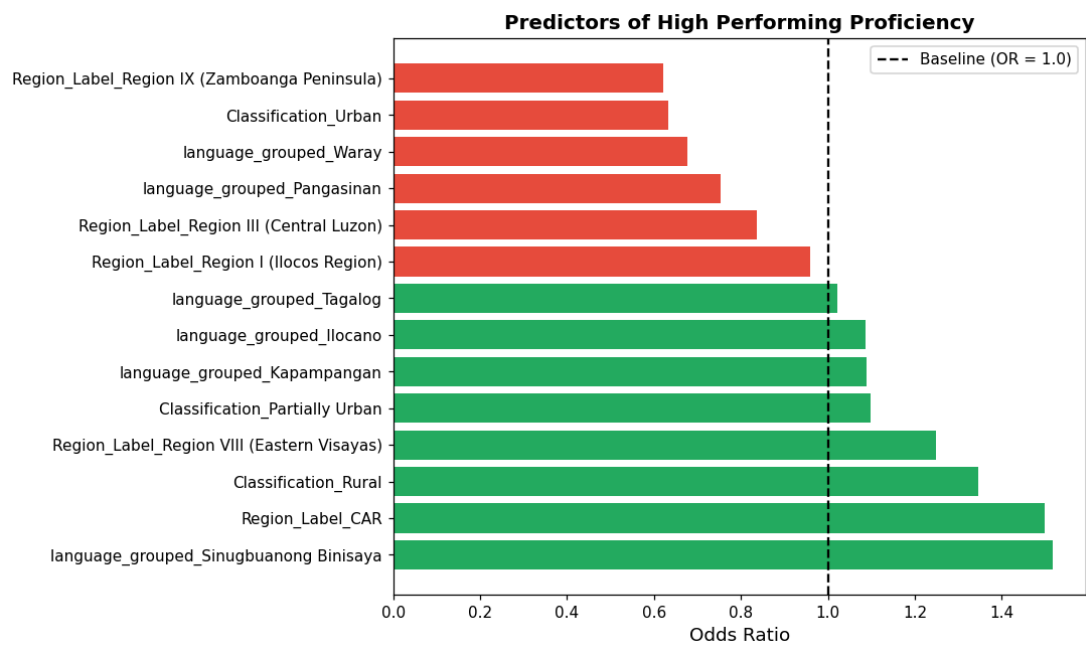


Figure 18. Predictors of High-Performing (Low Struggling-Reader) Schools

The most surprising result of the study is that, after Region and Mother Tongue are accounted for, Urban schools are actually less likely to be a high-performing, low-struggling school than Rural (OR = 0.6322) or Partially Urban (OR = 1.0973) schools. This is consistent with the finding at the whole sample level, reported in Table 11, that the Region III/Region VIII reversal is not an isolated result for Urban schools, but occurs at the whole sample level. Yet, despite Region VIII having some of the weakest At Grade Level rates in Table 9, Region also increases the likelihood of a high performer on this indicator (OR = 1.4992), as does Region VIII (OR = 1.2484). Region IX, with the highest At Grade Level rate in the model, is negatively associated with being a high performer on this indicator (OR = 0.6218). What is considered “high-performing” here is not as broad as it sounds, because if it were, then no child would ever be considered “at grade level” in the middle of the distribution, but rather it is a measurement of not having many struggling readers at the bottom of the distribution; it is a different question than being considered “at grade level” at the top of the distribution, and Region VIII and Region IX's performance in Model 2 and Model 3 below illustrate this distinction. The Urban result is counterintuitive, but the recent comments on the CRLA program itself have raised concerns that, once local implementation quality is considered, there is “no urban advantage.” The Urban result is



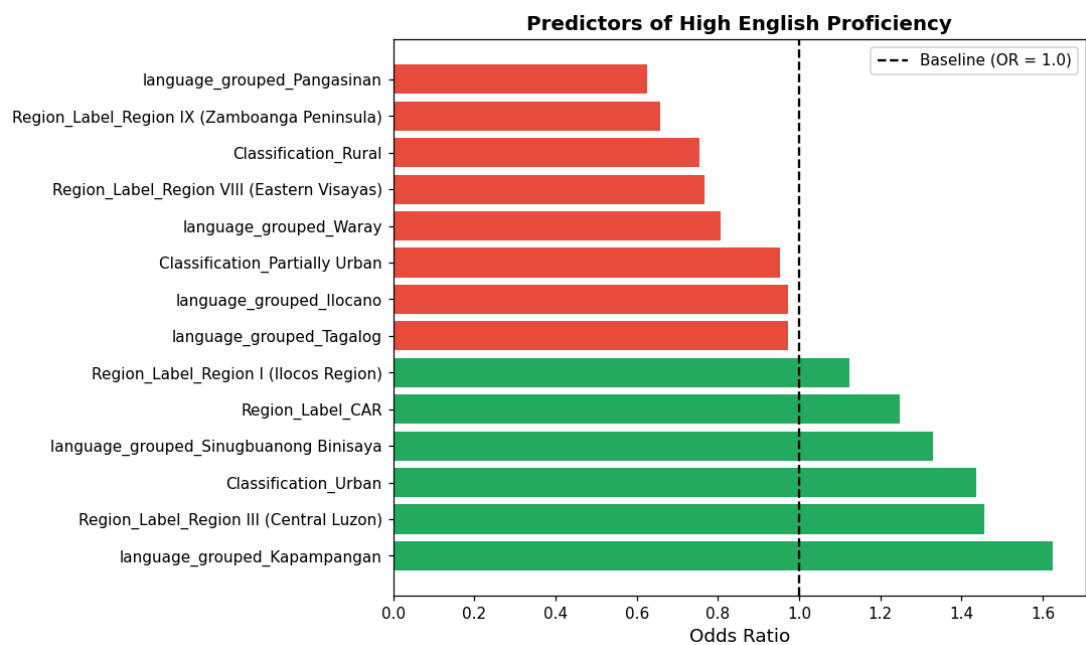
counterintuitive, and the recent comments about the CRLA program itself have raised concerns that there is "no urban advantage" once local implementation quality is taken into account (Philippine Star, 2026).

Determinants of High Grade 3 English Proficiency

Model 3 re-estimates the high-English-proficiency target from Model 1, this time with Region and Classification added alongside Mother Tongue, allowing the language effect observed in Model 1 to be checked against geography. Table 18 and Figure 15 report the results.

| Feature                    | Coefficient | Odds Ratio |
|----------------------------|-------------|------------|
| Kapampangan                | 0.4855      | 1.6251     |
| Region III (Central Luzon) | 0.3763      | 1.4568     |
| Classification: Urban      | 0.3621      | 1.4364     |
| Sinugbuanong Binisaya      | 0.2848      | 1.3295     |
| CAR                        | 0.2219      | 1.2484     |
| Region I (Ilocos Region)   | 0.1175      | 1.1247     |
| Tagalog                    | -0.0271     | 0.9732     |
| Ilocano                    | -0.0280     | 0.9724     |
| Classification: Partially  | -0.0475     | 0.9536     |
| Urban                      |             |            |
| Waray                      | -0.2134     | 0.8078     |
| Region VIII (Eastern       | -0.2650     | 0.7672     |
| Visayas)                   |             |            |
| Classification: Rural      | -0.2814     | 0.7548     |
| Region IX (Zamboanga       | -0.4175     | 0.6587     |
| Peninsula)                 |             |            |
| Pangasinan                 | -0.4686     | 0.6259     |

Table 20. Strongest Predictors of High Grade 3 English Proficiency



**Figure 19. Predictors of High Grade 3 English Proficiency (Model 3, Full Model)**

Model 3 shows that Kapampangan language speakers' higher achievement in high English language skills ( $OR = 1.6251$ ) are due to a convergence not only of linguistic benefits but also, and significantly, with regional resource benefits. The regional variable 'Region III' is an independent predictor of success ( $OR = 1.4568$ ), indicating that relatively stronger support systems through better Local Government Unit (LGU) and Special Education Fund (SEF) allocation in the region provide an enabling environment for Mother Tongue-Based Multilingual Education (MTB-MLE) to effectively serve as a scaffolding process for acquiring English proficiency (Second Congressional Commission on Education, EDCOM II, 2024). Recent studies indeed found out that by giving opportunities for learners to use their mother tongue as an intermediate stage for learning to write, basic literacy concepts can be transferred and influence the higher levels of the learners' English language skills (Deniega, 2024). Finally, the positive urban-rural effects of the classification of location ( $OR = 1.4364$ ) are expected as the urban-rural divide has long been documented in Philippine education, and that the literacy and English competence in schools located in urban areas is usually higher due to increased access to better learning materials, facilities, and quality teachers (UNESCO, 2023). In contrast, the negative predictive factors, classification of location ( $OR = 0.7548$ ), Region IX ( $OR = 0.6587$ ), and Pangasinan ( $OR = 0.6259$ ), reflect a systemic disparity of some parts of the country which lack



such capabilities. New research shows that differences in school facilities and reading materials severely impede literacy achievement in public elementary schools in the countryside, widen this urban-rural gap (Santos & Llenada, 2024; Wong, 2025), therefore proving that a policy needs to consider the geographical features, social economic condition and the language of instruction for specific policy interventions.

Determinants of High Grade 3 Filipino Proficiency

Model 4 repeats the full-model specification for high Grade 3 Filipino proficiency, using the same 20% At Grade Level threshold. Table 19 and Figure 16 report the results.

| Feature                         | Coefficient | Odds Ratio |
|---------------------------------|-------------|------------|
| Region III (Central Luzon)      | 0.6790      | 1.9719     |
| Pangasinan                      | 0.2993      | 1.3490     |
| Kapampangan                     | 0.1721      | 1.1878     |
| Classification: Partially Urban | 0.0669      | 1.0692     |
| Classification: Urban           | 0.0269      | 1.0273     |
| Region I (Ilocos Region)        | −0.0218     | 0.9785     |
| Sinugbuanong Binisaya           | −0.0511     | 0.9501     |
| Classification: Rural           | −0.0711     | 0.9314     |
| Region IX (Zamboanga Peninsula) | −0.0882     | 0.9156     |
| Ilocano                         | −0.0961     | 0.9084     |
| Tagalog                         | −0.1364     | 0.8725     |
| Waray                           | −0.1651     | 0.8478     |
| Region VIII (Eastern Visayas)   | −0.2236     | 0.7996     |
| CAR                             | −0.3227     | 0.7242     |

Table 21. Strongest Predictors of High Grade 3 Filipino Proficiency (Model 4, Full Model)

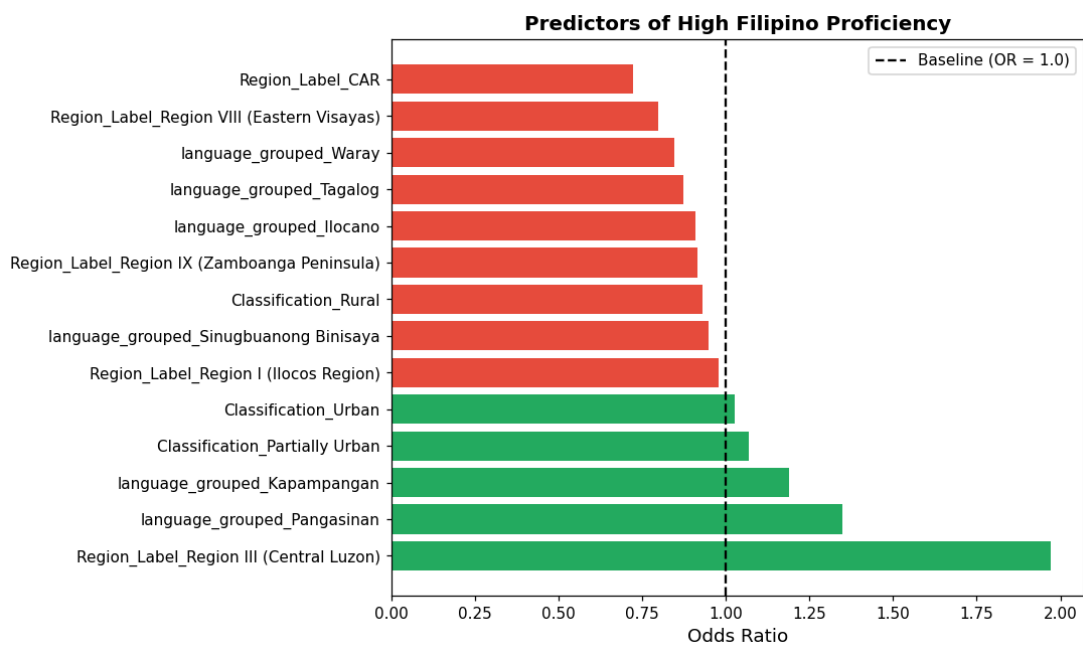


Figure 20. Predictors of High Grade 3 Filipino Proficiency (Model 4, Full Model)

Region III's advantage is strongest of all in the Filipino model (OR = 1.9719), nearly double the odds of the reference group, and considerably larger than its effect on English in Model 3 (OR = 1.4568). This is consistent with the correlation results in **Table 13**, where Mother Tongue correlated more strongly with Filipino ( $r = .9162$ ) than with English ( $r = .8074$ ): whatever drives Region III's outsized reading advantage appears to transfer more completely into the linguistically closer L2 than into the more distant L3. CAR shows the opposite pattern from its Model 3 result, here posting the weakest odds ratio of any predictor (OR = 0.7242), a reminder that a region's relative standing is not fixed across outcomes but depends on which language and which threshold of success is being modeled.

The boundaries of these demographic models demonstrate that background is not destiny. Only by examining the school building is it possible to determine if a program will be successful, as success is not necessarily determined by region and language of the student. An example of this is the strong performance of Central Luzon which may just be due to the size of its campuses, the ability to pool resources and to recruit specialized reading teachers (Kilag et al., 2024).



|                                | Nearly<br>Proficient | Not<br>Proficient | Proficient |
|--------------------------------|----------------------|-------------------|------------|
| Level_Purely ES                | 0.023597             | -0.941727         | 0.918130   |
| region_Region IX               | -0.069509            | 0.933654          | -0.864145  |
| region_Region VIII             | -0.050444            | -0.874582         | 0.925026   |
| language_Sambal                | 0.716432             | -0.175794         | -0.540638  |
| language_Kapampangan           | 0.704212             | -0.013539         | -0.690673  |
| Level_All Offering (K to 12)   | -0.225964            | 0.665928          | -0.439965  |
| language_Waray                 | -0.327922            | 0.631854          | -0.303932  |
| Level_ES and JHS (K to 10)     | 0.316073             | 0.313866          | -0.629939  |
| region_CAR                     | 0.568546             | -0.128837         | -0.439710  |
| region_Region III              | -0.254103            | -0.254838         | 0.508941   |
| language_Ilocano               | -0.037159            | -0.444114         | 0.481273   |
| n_test_takers_x                | -0.048049            | 0.429919          | -0.381870  |
| region_Region I                | -0.080784            | 0.362670          | -0.281887  |
| language_Yakan                 | -0.300154            | -0.045289         | 0.345444   |
| language_Southern Sorsoganon   | -0.289347            | -0.020020         | 0.309367   |
| language_Other                 | -0.080864            | -0.183968         | 0.264832   |
| language_Sinugbuanong Binisaya | -0.126051            | -0.134695         | 0.260747   |



|                                |           |           |           |
|--------------------------------|-----------|-----------|-----------|
| total_facilities               | -0.033023 | -0.217232 | 0.250255  |
| language_Pangasinan            | 0.081892  | 0.156607  | -0.238499 |
| language_Surigaonon            | -0.229182 | 0.105585  | 0.123597  |
| total_enrollees                | 0.065183  | -0.198198 | 0.133015  |
| Classification_Rural           | 0.184061  | -0.038394 | -0.145668 |
| school_low_emerging_percent    | -0.035304 | 0.170062  | -0.134758 |
| language_Tagalog               | 0.001850  | 0.161441  | -0.163290 |
| total_teachers                 | 0.117636  | -0.152543 | 0.034908  |
| Classification_Urban           | -0.073886 | 0.117457  | -0.043571 |
| school_at_grade_level_percent  | 0.007183  | -0.095739 | 0.088556  |
| school_transitioning_percent   | 0.019600  | -0.092396 | 0.072796  |
| school_developing_percent      | 0.022552  | -0.055370 | 0.032818  |
| Classification_Partially Urban | 0.003531  | -0.040996 | 0.037465  |
| school_high_emerging_percent   | -0.003682 | 0.040538  | -0.036856 |

**Table 22. Predictors of Multi-Class Logistic Regression Model**

Table 22 shows the coefficients from the multi-class logistic regression model that predicts whether a school is Not Proficient, Nearly Proficient, or Proficient based on overall\_mps, by including the predictors: region, language (mother tongue), school level, classification, and several size/resource indicators. An interesting technical characteristic is that for each predictor, the three coefficients sum to roughly zero, reflecting a sum-to-zero contrast coding scheme. This coding means each coefficient reflects the relative pull of the predictor



toward one performance category, relative to the average of all three categories, rather than relative to a removed category.

The regional effects are some of the most salient and generally conform with the CRLA descriptive results. Region IX (Zamboanga Peninsula) demonstrates the strongest pulling effect towards Not Proficient (0.93), and a substantial negative effect pulling away from Proficient (0.86). This result aligns with previous findings demonstrating regional disparities in the development of early grade literacy across the Philippines; with Mindanao regions often facing challenges related to diversity of language, resources and geographical remoteness (EDCOM 2, 2026; Balingbing & Tudlasan, 2025). In contrast, region III (Central Luzon) is strongly attracted to Proficient (0.51) and repulsed from Not Proficient (0.26), confirming its reputation as a stronger performing region. Interestingly, Region VIII (Eastern Visayas) is most attracted to Proficient (0.925), which is different from its descriptive profile which was weaker, suggesting an interaction effect with school level and/or mother tongue and or different thresholds being applied for “proficient” vs raw CRLA proficiency groups. This validates the work of Balingbing and Tudlasan (2025), who had mentioned that geographical isolation and limited resource availability in Mindanao have always been impediments to fundamental literacy development, and that the Region IX challenge may thus be more of a systemic, rather than an isolated one. Also, contrary to the assertion that the performance at regional levels is homogenous, this research has pointed out that Region VIII's unusual pull into the "Proficient" category at given thresholds indicated the intricate school-level/mother tongue interactions which were not discussed in previous descriptive reports.

Similarly, the language/mother tongue categories show relevant patterns, with Sambal and Kapampangan attracting to Nearly Proficient, and Waray to Not Proficient (0.63). The languages that cluster in the direction of Proficient are numerous, such as Ilocano, Sinugbuanong Binisaya and “Other”. These are likely indicative of the country's experience with MTB MLE with the different language groups, as certain language groups perform better and others perform poorer with MTB MLE (Igarashi et al., 2024; Reyes & Cruz, 2025). Tagalog is relatively neutral, probably because of its use in a relatively large amount and its relatively high resources. This



validates the work of Igarashi et al. (2024) who previously posited that the efficiency of MTB-MLE relies greatly on the provision of localized learning materials-which for languages such as Waray, are rare.

The characteristics of the school also influence. Purely Elementary Schools have one of the greatest pulls towards Proficient (0.92) and away from Not Proficient (0.94), which may be caused by focused efforts in this grade. Having a wider range of offerings (ES + JHS or K-12) draws to lower levels of performance. Resources may be focused on early grades only at purely elementary levels to promote achievement. This validates the work of Kilag et al. (2024) who pointed out that schools concentrated on smaller grades can better channel more time and resources to the foundational learning of the earlier grades, as opposed to a school which spreads its focus and resources thinly across several grade levels. Unlike typical notions of an urban vs. Rural distinction, this study has suggested that being in the city cannot directly equate to higher outcomes in the long run when considering controlling for certain structural issues.

When structural factors are taken care of, learning elementary skills is not necessarily an urban advantage; slight performance toward Nearly Proficient in rural areas and toward Not Proficient in urban areas is indicative of being in the city (Miral & Santos, 2025; EDCOM 2, 2026). Not Proficient (0.43) is the value of the variable `n_test_takers_x` indicating that the scale of operation may have an impact on the results. The total facilities, totalteachers, totalenrollees have little, varied effect in this analysis. Also, note that the school proficiency percentages (`low_emerging`, `high_emerging`, etc.) were not included since they were likely to be collinear with the dependent variable (school achievement). This is also reflected on the comments of Miral and Santos (2025) who noted that the "urban advantage" may be a cover effect because of the phenomenon of overcrowding and diluted resources in urban schools, while rural schools are better prepared to offer context-determined teaching because of their increased capacity for context-specific approaches based on the local community.

Therefore, these variables represent a combined total of 100 percent, it's plausible these overlap considerably with the categorical boundaries that were drawn around the outcome variable categories. Other avenues of future modeling might investigate the viability of including



these variables in other ways - through different estimation techniques, transformation, or others in order to alleviate problems with multicollinearity. This pedagogical point supports the observations of Orbeta et al. (2025) that current predictive modeling approaches on multilingual education tend to have an accuracy cap because they rely on proxy variables such as demographics and a more granular classroom data needs to be collected for future research.

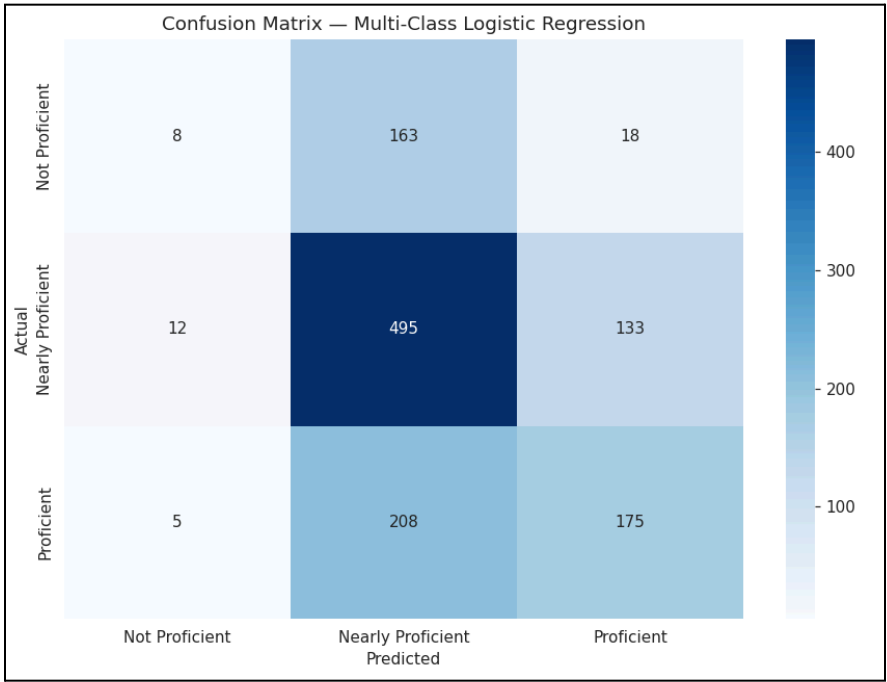
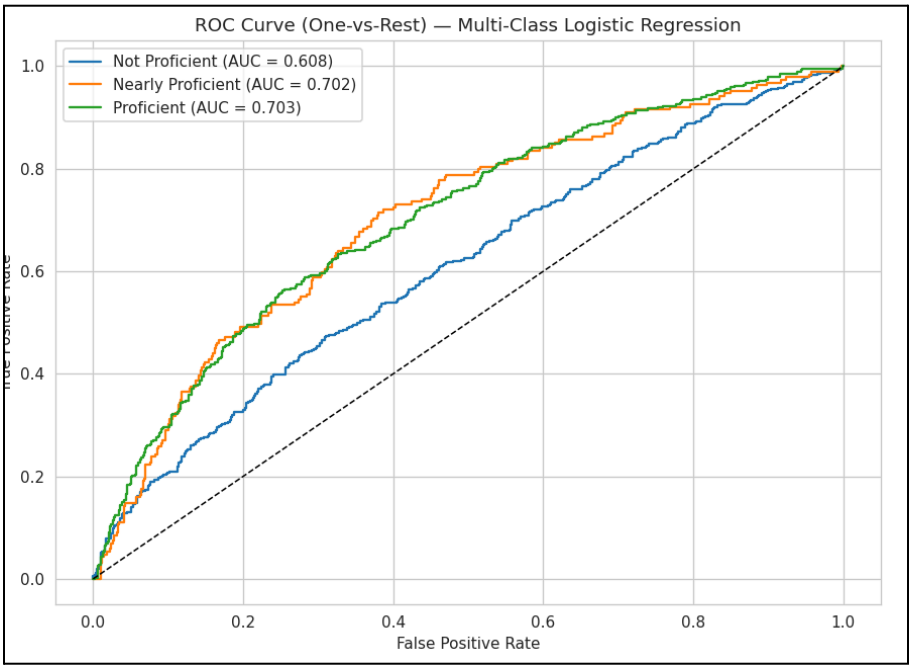


Figure 21. Confusion Matrix of the Multi-Class Logistic Regression Model

The confusion matrix indicates a reasonable ability to correctly classify "Nearly Proficient" school predictions (495 correct). But the extremes are a big problem, it missed 91 out of 114 "Not Proficient" cases, and missed 8 out of 189 cases in that level. Many were incorrectly classified as "Nearly Proficient". This is the trend in Filipino schools. The model is more likely to find a safe middle ground, a situation that reflects reality of the literacy outcomes of school which are either not failing or not successful despite the problems or issues in the system (William et al., 2025).

The constraints are based on the results of recent studies regarding the education situation in the Philippines. The accuracy of predicting literacy levels based on demographic indicators is relatively low because other factors such as the quality of learning and teaching and gaps in resources have a greater effect. The need for a policy that is not only language-centric but also addresses real world problems is highlighted (Orbeta et al., 2025) .



**Figure 22. ROC Curve of the Multi-Class Logistic Regression Model**

ROC curves in Figure 22 provide a more detailed picture of the merits and drawbacks of the model. AUC scores are around 0.70 for "Nearly Proficient" and "Proficient" classes, but fall to 0.608 for "Not Proficient." This implies that the model only has a certain level of ability of discriminating between average-to-good performers and the rest, but cannot do very much for the very weak schools where teachers and educational communities are most concerned.

This limited capability is in line with the general work on predictive modelling in multilingual education, which pinpoints general trends and fails to capture edge cases due to the complexity of socio-linguistic factors (SHARE LITES Report, 2026). It serves as a reminder that data mining can be used to gain insights, but human-centred support is essential for learners who struggle in learning (Orbeta et al., 2025) .



|                   | Precision | Recall | F1-score | Support |
|-------------------|-----------|--------|----------|---------|
| Nearly Proficient | 0.57      | 0.77   | 0.66     | 640     |
| Not Proficient    | 0.32      | 0.04   | 0.07     | 189     |
| Proficient        | 0.54      | 0.45   | 0.49     | 388     |
| accuracy          |           |        | 0.65     | 1178    |
| Macro avg         | 0.68      | 0.62   | 0.41     | 1178    |
| Weighted avg      | 0.52      | 0.56   | 0.61     | 1178    |

Table 23. Multi-class Regression Evaluation

The overall performance of the model is now summarized in Table 23, which has an accuracy of 65%, good recall for “Nearly Proficient” schools (0.77), and alarmingly low recall for “Not Proficient” schools (0.04). It will be common knowledge among education players in the Philippines that schools are mediocre at best, and that underperforming schools are more difficult to pinpoint and address with a general data tool.

This is similar to the findings of recent assessments of predictive models in teacher licensure and literacy transitions, where a limited set of features were found to yield similar accuracy limits (Patungan, 2025) ; (Orbeta et al., 2025) . The final deliverables emphasize the need to adopt a hybrid model that leverages analytics and place-based data to have a positive impact on student outcomes.



## **CONCLUSION**

This study sought to determine the effect of the use of the Mother Tongue (L1) on L2 (Filipino) and L3 (English) proficiencies using school-level data from the CRLA 2025-2026 dataset. The sample covered 6,121 public elementary schools and 647,681 learners from 5 regions during the MTB-MLE program implementation period. Furthermore, the study also examined the change in association between the effects of L1 proficiency and L2/ L3 outcomes across regions and school types, and used regression modeling to determine the predicting power of Mother Tongue proficiency in L2/L3 performance.

Results revealed that there is a highly significant relationship between Mother Tongue, Filipino, and English up to Grade 3 level where the most significant association exists between Mother Tongue and Filipino ( $r = .9162$ ), followed by Filipino and English ( $r = .8325$ ) and Mother Tongue and English ( $r = .8074$ ). Consequently, the results led to rejection of the null hypothesis, supporting the assertion that schools which have high proficiency in Mother Tongue tend to perform well in Filipino and English, consistent with Cummins' (1979) linguistic interdependence hypothesis.

Regional variations also became evident. Region III (Central Luzon) has the highest At Grade Level proficiency (18.40%) while Region VIII (Eastern Visayas) has the lowest At Grade Level proficiency (12.69%). Moreover, urban schools performed better than rural schools with few exceptions as Urban schools in Region VIII did worse than Rural schools within the same region showing that location supersedes urban school characteristics.

There was a principal component analysis wherein the first principal component (PC1) is "overall reading proficiency" which is responsible for almost 62% of the variance in the total variation among schools, indicating that the difference among schools could be attributed to the overall proficiency of their reading programs in different languages and grade levels. On the other hand, PC2 focuses on schools whose early proficiency cannot carry over to grade 3 level.

The predictive models showed that while Mother Tongue, Region, and Classification are statistically detectable predictors, their predictive power is modest, with accuracy ranging from 54.41% to 61.44%. This suggests that demographic and linguistic factors aren't the only factors



that determine school-level success; as what happens inside the school, like teacher quality, instructional materials, and sustained interventions matters just as much as the language of instruction.



## **RECOMMENDATIONS**

Building upon the observations reported in the previous discussion, the following recommendations are put forth to policymakers, school administrators, and the research community:

**To Policymakers and Legislators.** The significant positive correlations found among the three languages lend empirical observational support for the cross-linguistic transfer assumption that was originally the primary justification for Republic Act No. 10533. These results caution against reading Republic Act No. 12027 as a refutation of language theory, as opposed to its problematic implementation. Before applying the findings of this research to future language-in-education policy, it is recommended that legislators and DepEd officials commission an implementation-fidelity audit covering both teacher training completion and learning material availability and accuracy in language mapping. Without this audit, this analysis cannot differentiate the "true" mother-tongue effect from the concomitant region-level resource advantage documented throughout the Results and Discussion. Where Republic Act No. 12027 may allow for optional use of Mother Tongue as an auxiliary instructional language in selected monolingual classrooms, this study recommends that DepEd implement this auxiliary language support in regions with the highest aggregate proportion of struggling readers, which are the Region VIII and Region IX, rather than attempt a nation-wide approach. Targeted regional resource allocation strategies, rather than uniform national ones, have been repeatedly shown to be most effective in mitigating regional literacy differences (Second Congressional Commission on Education [EDCOM II], 2024).

**To the Department of Education and School Administrators.** The concentration of deficiency in Region IX among learners stalled at Low Emerging differs substantially from that of Region VIII, among learners in that region who very rarely attain Grade Level while having a relatively low Low Emerging score; accordingly, a nationalized, one-size-fits-all intervention program for all learners is very unlikely to be effective in either region.



Foundational decoding and phonics based instruction, to overcome the initial stages of reading development, would be most effective for the vast majority of deficient students in Region IX. In contrast, advanced reading comprehension, to bridge the Transitioning band to Grade Level proficiency, would more appropriately address the deficits observed in Region VIII. Evidence-based, differentiated reading instruction is indispensable to addressing foundational learning poverty (World Bank, 2023). Because Urban status fails to uniformly predict stronger academic outcomes with some Urban schools reporting the lowest Grade Level rates among any subgroup this study recommends a shift in resource allocation strategy from purely an urban-rural distinction toward a needs-based approach that integrates proficiency data with school classification. Resource planning for Rural schools may also more appropriately target upper-primary reading interventions and reading specialists as opposed to assuming a primary need for foundational literacy education, due to the fact that they tend to cluster disproportionately in the Transitioning rather than the Low Emerging band. In the case of the Cordillera Administrative Region, with an unusually high concentration of rural schools owing to its mountainous geography, specific consideration should be made in learning material procurement and teacher deployment plans to account for these logistical challenges.

**To Teacher Training and Instructional Materials Development.** While the odds of high English proficiency were significantly highest in Kapampangan-medium schools when solely controlling for Mother Tongue, the odds declined once Region III was added as a covariate suggesting the language advantage may reflect Region III's broader resource endowment rather than an independent language effect.

Before implementing language-specific teacher training or instructional materials based on these language-based odds ratios, it is recommended that DepEd Commission region-controlled follow-up analyses that can disaggregate genuine language effects from correlated regional ones.

Meanwhile, in light of already documented issues with shortages in instructional materials and teacher trained to deliver them, teacher training investment should be prioritized in Waray- and



Pangasinan-medium schools, which displayed the weakest odds of high English proficiency regardless of whether or not region was entered into the models.

**To Future Researchers.** Because household income, nutrition, and health were not captured in this study, future research should explicitly include these socioeconomic covariates, as has been called for by others (Kilag et al., 2024) and supported by recent national data (Philippine Statistics Authority [PSA] & UNICEF, 2023) to assess their contribution to these strong positive correlations. Due to the modest accuracy of the predictive models in this study, future work on building these models might consider incorporating additional rich school-level data, including measures of teacher to student ratios, teacher training completion rates, and textbooks available per learner; since the Mother Tongue, Region, and Classification labels only produced modest detectable signal. Future studies comparing outcomes of learner cohorts prior to and subsequent to the implementation of Republic Act No. 12027 could explore the causal effects of removing the mandatory mother-tongue instruction through the use of a longitudinal or quasi-experimental design. This study also recommends continuing the exploration of learning trajectories through the full hierarchical clustering analysis, outlined in the Methodology section, that sought to identify natural school profiles beyond simply the demographic markers studied here.



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## APPENDICES

**Dataset CSV Link:**

[https://drive.google.com/file/d/1xjUH3fHpm37YNfhEKWnGi0KnrodliOzA/view?usp=drive\\_link](https://drive.google.com/file/d/1xjUH3fHpm37YNfhEKWnGi0KnrodliOzA/view?usp=drive_link)

**Google Sheet Link:**

<https://docs.google.com/spreadsheets/d/1HqVQ3HCkQaqRbx5Lelk4OPiKcQn3S17m0QRdBEMT7jo/edit?usp=sharing>

**Google Colab Link for Descriptive Statistics and Hierarchical Clustering:**

<https://colab.research.google.com/drive/1UmNLsT4uiMR8LrMGmwfgsqwEJhbN6uWc?usp=sharing>

**Google Colab Link for Multi-Class Logistic Regression:**

<https://colab.research.google.com/drive/1CU9ApeMGCzcmPCp4atiRGDzHU7K09dlf?usp=sharing>

**CRLA 2025-2026, Early Language Literacy and Numeracy Assessment (ELLNA), School Facilities in SY 2023-2024, Enrollment in SY 2024-2025, School Personnel in SY 2023-2024**

**Csv Link:**

<https://www.deped.gov.ph/machine-ready-files/>

**Canva Link:**

<https://canva.link/89eklzbxepqmu8s>

**Turnitin Similarity Report Link:**

[https://ev.turnitin.com/app/carta/en\\_us/?o=2998752429&s=1&ro=103&student\\_user=1&lang=en\\_us&u=1201500998](https://ev.turnitin.com/app/carta/en_us/?o=2998752429&s=1&ro=103&student_user=1&lang=en_us&u=1201500998)